STRAIGHTENING LAWS IN ALGEBRAIC COMPLEXITY

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Abstract. Algebraic complexity theory studies the difficulty of computing certain polynomials in various models of computation. The most natural form of computation is that of the algebraic circuit. Proving lower bounds on the complexity of a polynomial turns out to be much more difficult, and we have very few examples. Rather than asking about the complexity of a fixed polynomial such as the $n \times n$ determinant, one can ask about proving lower bounds of the complexity of a polynomial ideal. It is currently not known how to prove lower bounds for general polynomial ideals. However, we can study ideals where we have significantly more structure. Determinantal ideals, generated by minors of a matrix, have additional structure which allow us to have a finer control when manipulating elements of these ideals. This additional structure comes in the form of a straightening law. With this additional structure, we can prove lower bounds for determinantal ideals, and use similar techniques for other ideals with straightening laws.

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1. Introduction

Many problems in combinatorics, geometry, and computer science can be expressed in terms of the computation of some polynomials. For example, determining if a graph has a perfect matching is equivalent to determining if the determinant of it's *Tutte Matrix* is identically zero [Tut47]. With this, we see that it is important to study the complexity of computing polynomials in general. Specifically, we study the difficulty of computing special families of polynomials such as the determinant or permanent polynomials. Algebraic complexity theory is the study of the complexity of such computations.

1.1. **Algebraic Complexity Theory.**

Studying the optimality of the number of operations to compute a given polynomial is one of the core problems of algebraic complexity. This is known as the *minimal circuit complexity* problem, which by circuit we mean an *algebraic circuit* (Definition 2.1). Perhaps the earliest work in this direction was a proof of optimality of *Horner's Rule* where Ostrowski showed that one cannot compute a generic degree-d univariate polynomial using fewer than d additions [Ost54]. Strassen showed that one can multiply matrices faster than the usual algorithm [Str69], giving an improvement from $O\left(n^3\right)$ to $O\left(n^{\log_2 7}\right)$. Csanky give fast parallel algorithms for various problems in linear algebra, in particular giving a fast method for computing the determinant [Csa76]. Proving upper bounds on the computational complexity of certain polynomials comes down to devising new algorithms for them. However, proving lower bounds is typically much harder as one must rule out the possibility of all possible algorithms. This often comes down to devising new complexity measures and invariants, and characterizing exactly which polynomials are "small" with respect to these measures. See the survey of Saptharishi for further details in this direction [Sap21].

Another important problem in algebraic complexity is the problem of deciding if a given algebraic identity is identically zero. This is known as the *polynomial identity testing* problem. A fast randomized algorithm for this problem over arbitrary fields was independently discovered by Schwartz and Zippel [Sch80; Zip79]. A version for finite fields was given earlier by Ore in 1922 [Ore22]. Derandomizing algorithms for polynomial identity testing is deeply connected to proving lower bounds on minimal circuit size [NW94; KI04]. However, for general algebraic circuits, deterministic algorithms which are an improvement of the naive derandomization of the Schwartz-Zippel algorithm are not known. Any subexponential deterministic polynomial identity testing algorithm for general circuits would yield highly nontrivial circuit complexity lower bounds. For further progress on derandomizing polynomial identity testing, the survey of Dutta and Lysikov is a great reference [DL25].

In computational complexity, one tries to characterize the complexity of a given problem and determine which *complexity class* that problem lives in. For example, the class P is the set of decision problems, yes and no problems, whose solution can be computed in polynomial time. A closely related class is NP, the set of decision problems whose solution can be *verified* in polynomial time. The question of separating P and NP is the largest open problem in computer science. To do this, one strategy is to find so called *complete* problems in the class NP, and show that these problems are not contained in P. Complete problems in a class such as NP have the property that every problem instance in NP can be efficiently transformed into an instance of the complete problem. Thus, to solve any problem in NP in polynomial time, it suffices to solve a complete problem in polynomial time. For example, the question of finding a satisfying assignment for a boolean formula is NP-complete. Thus, every problem in NP can be efficiently transformed into a boolean formula which is satisfiable if and only if the original problem has a yes solution. This process is known as *reducing* to a complete problem.

In two seminal works [Val79; Val81], Valient defined algebraic analogues of complexity classes, analogues of complete problems for these complexity classes, and a notion of reduction. Valient's algebraic analogue of P, known as VP, can be described as n-variate polynomials with $\mathrm{poly}(n)$-degree which are efficiently computable. The analogue of NP, known as VNP, can be described as n-variate polynomials with $\mathrm{poly}(n)$-degree whose *coefficients* are efficiently computable. See Definition 2.13 and Definition 2.16 for formal statements. One complete polynomial for VNP is the $n \times n$ permanent. While the determinant is known to be efficiently computable in many standard algebraic models of computation [Csa76], it is not known if the permanent can be efficiently computed. In fact, it is conjecured that this is not the case.

Conjecture 1.1: Let $\mathbb{F}$ be a field of characteristic different from two, $X = (x_{i,j})$ be an $n \times n$ matrix of variables and $m, n \in \mathbb{N}$ such that $m \geq n$ Suppose that A is a matrix of polynomials of degree at most 1 over $\mathbb{F}$ such that $\det_m(A) = \mathrm{perm}_n(X)$. Then, we cannot have that $m = \mathrm{poly}(n)$, m must be superpolynomial in n.

Already here we see that field independent methods cannot resolve Conjecture 1.1 since over fields of characteristic two, the determinant and permanent are the same polynomial.

1.2. **Straightening Laws.** A natural extension of studying the complexity of the determinant polynomial is to study the complexity of *determinantal ideals*. These are the ideals generated by the minors of a matrix of variables. One feature of these ideals is that they admit an alternative basis to the monomial basis, and importantly this basis is described combinatorially. With such a basis, we get more control over the cancellations that may occur as well. Specifically, we get a *straightening law*, which describes how to derive expressions in this combinatorial basis. The study of such ideals, and their coordinate rings, with straightening laws is known as *standard monomial theory*. One of the most classical examples of standard monomial theory and straightening laws comes from the coordinate ring of the Grassmannian, whose relations are the *(quadratic)-Plücker relations*. Many good references on this topic can be found [Ful96; Stu08; LB18].

Let V be an n-dimensional vector space over a field $\mathbb{F}$ with basis $\{e_1, \ldots, e_n\}$. We will notate by $\Lambda(n,d) \subseteq [n]^d$ the set of all ordered tuples $(i_1 < \cdots < i_d)$ where each $i_k \in [n]$.

Definition 1.2 (Grassmannian): Let $\mathrm{Gr}(n,d) = \{W \subseteq V \mid \dim_{\mathbb{F}}(W) = d\}$ be the d-dimensional *Grassmannian*. Equivalently, $\mathrm{Gr}(n,d)$ can be thought of as the quotient of full-rank $d \times n$ matrices by $\mathrm{GL}_d(\mathbb{F})$.

One can give this the structure of an algebraic variety via the following map.

Definition 1.3 (Plücker Embedding, Plücker Coordinates): For $U \in \mathrm{Gr}(n,d)$, let U have basis $\{u_1, \ldots, u_d\}$. Define a map $\mathrm{Gr}(n,d) \to \mathbb{P}\left(\bigwedge^d V\right)$ given by $U \mapsto [u_1 \wedge \cdots \wedge u_d]$. This map is called the *Plücker embedding*.

The set $\{e_{\bar{i}} = e_{i_1} \wedge \cdots \wedge e_{i_d} \mid \bar{i} \in \Lambda(n,d)\}$ gives a basis for $\bigwedge^d V$. Let $\{p_{\bar{j}}\}_{\bar{j} \in \Lambda(n,d)}$, denote the basis of $\left(\bigwedge^d V\right)^*$ dual to $\{e_{\bar{i}}\}_{\bar{i} \in \Lambda(n,d)}$. Then for each $U \in \mathrm{Gr}(n,d)$, we can represent U via an $n \times d$ matrix of rank d via the basis $\{u_1, \ldots, u_d\}$. Then, $p_{\bar{j}}(U)$ is the $d \times d$ determinant of the minor defined by rows $\bar{j} \in \Lambda(n,d)$. The coordinates $p_{\bar{j}}$ are known as *Plücker coordinates*. We will often identify the Plücker coordinate $p_{j_1,\ldots,j_d}$ with the associated determinant.

Definition 1.4 (Plücker Relations): For $(i_1, \ldots, i_{d-1}) \in \Lambda(n, d-1)$ and $(j_1, \ldots, j_{d+1}) \in \Lambda(n, d+1)$, define the following quadratic polynomial in the Plücker coordinates:

$$\sum_{i=1}^{d+1} p_{i_1,\ldots,i_{d-1},j_i} \cdot p_{j_1,\ldots,\widehat{j_i},\ldots,j_{d+1}}.$$

The ideal $\mathfrak{q}$ of these quadratic polynomials is the ideal of *Plücker relations*.

Proposition 1.5 ([LB18, Theorem 12.1.3] and [Ful96, §8.4, Proposition 2]): The zero set of the ideal $\mathfrak{q} \subseteq \mathbb{F}[p_{\bar{j}}]$ of quadratic relations is the Grassmannian $\mathrm{Gr}(n,d)$. Furthermore, $\mathfrak{q}$ is prime so that the $\mathrm{Gr}(n,d)$ is irreducible.

Proposition 1.6: The dimension of $\mathrm{Gr}(n,d)$ is $d(n-d)$.

Proof: To describe a d-dimensional subspace of $\mathbb{F}^n$, it suffices to give a $d \times n$ matrix M. Note that the set X of full-rank matrices is a Zariski open, and thus dense, subset and so we can assume such M are full-rank. This gives dn variables. Then, the dimension of $\mathrm{GL}_d(\mathbb{F})$ is d^2. We have a map $\pi : X \to \mathrm{Gr}(n,d)$ mapping a matrix to it's row span. The dimension of any of the fibers is d^2 so that $\dim(\mathrm{Gr}(n,d)) = dn - d^2 - d(n-d)$. □

Definition 1.7 (Plücker / Bracket Algebra): By Proposition 1.5, the coordinate ring $\mathbb{F}[\mathrm{Gr}(n,d)]$ of $\mathrm{Gr}(n,d)$ is given by

$$\mathbb{F}[p_{\bar{i}} \mid \bar{i} \in \Lambda(n,d)]/\mathfrak{q}.$$

In fact, one can also show that this coordinate ring is also the invariant subring $\mathbb{F}[x_{i,j} \mid 1 \le i \le n, 1 \le j \le d]^{\mathrm{SL}_d(\mathbb{F})}$. This algebra is known as the *Plücker* algebra or the *bracket algebra*.

The fact that the invariant subring $\mathbb{F}[x_{i,j}]^{\mathrm{SL}_d(\mathbb{F})}$ is generated by the $d \times d$ minors is known as the *First Fundamental Theorem of Invariant Theory*. That the relations between the minors are all generated by the Plücker relations is known as the *Second Fundamental Theorem of Invariant Theory*.

We can represent the product of k minors using a $d \times k$ tableau. One standard reference for this theory is [Stu08].

Definition 1.8 (Tableau, Tableaux Ordering, Standard Monomials): We can order the minors $p_{i_1,\ldots,i_d} \prec p_{j_1,\ldots,j_d}$ if there is some $1 \le m \le d$ such that $i_k = j_k$ for $1 \le k \le m-1$ and $i_m < j_m$. This gives a total order on the Plücker coordinates $p_{\bar{j}}$ which induces a degree reverse lexicographic monomial ordering on the bracket algebra $\mathbb{F}[p_{\bar{j}}]$. This ordering is known as the *tableaux ordering* as we can express a product of k coordinates $\bar{i}^1 \preceq \cdots \preceq \bar{i}^k$ as a tableau

$$\begin{bmatrix} i_i^1 & \cdots & i_d^1 \\ i_i^2 & \cdots & i_d^2 \\ & \vdots & \\ i_i^k & \cdots & i_d^k \end{bmatrix}.$$

We will say that such a monomial is *standard* if the columns are sorted (recall the rows are already sorted by convention). Otherwise, such a monomial is *nonstandard*.

This *standard monomial* theory is originally due to Hodge [Hod43]. However, to express the product of standard monomials in terms of other standard monomials, we use a *straightening* whose formulation is originally due to Young [You28]. Hodge used Young's formulation in terms of tableau in order to study the coordinate ring $\mathbb{F}[\mathrm{Gr}(n,d)]$.

Definition 1.9 (Van der Waerden Syzygies, Straightening Syzygies): For $\overline{\lambda} \in \Lambda(n,d)$, let $\overline{\lambda^*}$ be the *complement* tuple in $\Lambda(n,n-d)$ consisting of the $n-d$ elements which do not appear in $\overline{\lambda}$. The *sign* of $\overline{\lambda}$ and $\overline{\lambda^*}$, denoted by $\mathrm{sgn}(\overline{\lambda},\overline{\lambda^*})$, is the sign of the permutation $\sigma \in \mathfrak{S}_n$ given in one-line notation by $\lambda_1 \cdots \lambda_d \lambda_1^* \cdots \lambda_{n-d}^*$.

For $s \in [d]$, $\overline{\alpha} \in \Lambda(n, s-1)$, $\overline{\beta} \in \Lambda(n, d+1)$, and $\overline{\gamma} \in \Lambda(n, d-s)$, the *Van der Waerden syzygy* $\left[\left[\overline{\alpha}\dot{\overline{\beta}}\overline{\gamma}\right]\right]$ is the following quadratic polynomial in $\mathbb{F}[p_{\overline{i}} \mid \overline{i} \in \Lambda(n,d)]$:

$$\left[\left[\overline{\alpha}\dot{\overline{\beta}}\overline{\gamma}\right]\right] := \sum_{\overline{\tau} \in \Lambda(d+1,s)} \operatorname{sgn}(\overline{\tau}, \overline{\tau^*}) \cdot p_{\alpha_1,\ldots,\alpha_s,\beta_{\tau_1^*},\ldots,\beta_{\tau^*_{d+1-s}}} \cdot p_{\beta_{\tau_1},\ldots,\beta_{\tau_s},\gamma_1,\ldots,\gamma_{d-s}}.$$

Such a Van der Waerden syzygy is a *straightening syzygy* if $\alpha_{s-1} < \beta_{s+1}$ and $\beta_s < \gamma_s$. Note that if we take $s = 1$, we get the Plücker relations.

Theorem 1.10 ([Stu08, Theorem 3.1.7], [Rai]): The set of straightening syzygies is a Gröbner basis for the ideal of quadratic Plücker relations $\mathfrak{q}$ under the tableaux order. A monomial is standard if and only if it does not lie in the initial ideal $\textsc{lm}_{\preceq}\, \mathfrak{q}$. Thus, the set of standard monomials forms an $\mathbb{F}$-vector space basis for $\mathbb{F}[\mathrm{Gr}(n,d)]$. If we declare each $p_{\overline{i}}$ to have degree one, then the standard tableau of shape $d \times k$ give a basis for the degree k graded component of $\mathbb{F}[\mathrm{Gr}(n,d)]$.

Example 1.11: We work in $\mathbb{F}[\mathrm{Gr}(6,3)]$. Consider the tableau $T = [145][156][234]$. Then this is nonstandard in the second column as $5 > 3$. Thus, we reduce T by the syzygy

$$[[1\dot{5}\dot{6}\dot{2}34]] = [156][234] + [136][245] - [135][246] - [126][345] + [125][346] + [123][456].$$

Then, we have that

$$\begin{aligned} T - [145][[1\dot{5}\dot{6}\dot{2}34]] =& [136][145][245] + [135][145][246] \\ &+ [126][145][345] - [125][145][346] - [123][145][456]. \end{aligned}$$

Note that all tableau in the above expression are smaller than T with respect to the tableau order so that this process eventually terminates.

2. Preliminaries on Algebraic Complexity

2.1. **Computational Models.**

Definition 2.1 (Algebraic Circuits): An *algebraic circuit* over a field $\mathbb{F}$ is a directed acyclic graph whose *input nodes*, nodes of in-degree zero, are labeled either by variables $x_1, \ldots, x_n$ or constants from $\mathbb{F}$. Nodes of in-degree greater than zero are labeled by Σ or Π which are *addition gates* or *multiplication gates* respectively. Each edge or *wire* is also labeled by a constant from $\mathbb{F}$.

Computation is defined inductively node by node. Input nodes compute their label. Every Σ gate computes the sum of their children multiplied by the constants along each wire, and computation at Π gates is defined similarly. Nodes of out-degree zero are *outputs*. We say that an algebraic circuit *computes* $\{f_1(\overline{x}), \ldots, f_k(\overline{x})\}$ if these are the polynomials computed by all the outputs.

The *size* of a circuit is the total number of edges. We will denote by $size(f_1, \dots, f_k)$ the size of the smallest circuit over $\mathbb{F}$ computing $\{f_1(\overline{x}), \dots, f_k(\overline{x})\}$. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has $O\left(v^2\right)$ edges. It's *depth* is the length of the longest input-to-output path. The *fan-in* and *fan-out* of a node are in-degree and out-degree respectively. The *fan-in* and *fan-out* of the circuit is the maximum fan-in and fan-out respectively over all nodes in the circuit. The *degree* or *syntactic degree* of a circuit is the maximum degree of a polynomial computed by any of the nodes of the circuit.

Definition 2.2 (Algebraic Formulas): An *algebraic formula* over a field $\mathbb{F}$ is an algebraic circuit whose underlying graph is a tree.

Definition 2.3 (Constant Depth Circuits): Without loss of generality, we can assume that a circuit is made up of alternating layers of Σ and Π gates. We can use this to describe a restriction on the depth and types of gates appearing on each layer. For example

- A $\Sigma\Pi$ circuit computes a polynomial via it's monomial or *sparse* representation.
- A $\Pi\Sigma$ circuit computes a polynomial as a product of degree ≤ 1 factors.
- A $\Sigma\Lambda\Sigma$ circuit computes a polynomial as a sum of powers of degree ≤ 1 polynomials. This model is an inhomogeneous version of *Waring Rank*.

Definition 2.4 (Minimal Circuit Size Problem): Given an n-variate degree-d polynomial $f(x_1, \dots, x_n)$, find an algebraic circuit of minimal size computing $f(x_1, \dots, x_n)$.

Example 2.5: Let $f(x) = \sum_{i=0}^{d} a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Then *Horner's Rule* gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_0, \qquad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

This recurrence can be formulated as an algebraic circuit of size $O(d)$ and depth $O(d)$. In fact, this uses exactly d additions and d multiplications, which is optimal. The proof of optimality of the number of additions for computing a generic univariate polynomial $f(x)$ in this manner was given by [Ost54]. The corresponding proof of optimality for multiplications was given by [Pan66].

It turns out that "most" n-variate degree d polynomials require large circuits.

Proposition 2.6: Let $\mathbb{F}$ be a field of characteristic zero. Then most n-variate polynomials of degree d require circuits of size $\binom{n+d}{d}$.

Proof: First, the number of monomials in $\mathbb{F}[x_1, \dots, x_n]$ of degree at most d is $N := \binom{n+d}{d}$. Next, recall that algebraic circuits are described by directed acyclic graphs which have edge labels. An algebraic circuit with v nodes will have $s \leq \binom{v}{2} = \frac{v(v-1)}{2}$ edges. Fix a circuit topology C of size s with

v nodes which computes a polynomial of degree at most d over $\mathbb{F}[x_1,\dots,x_n]$. Then define a map $\Phi\colon \mathbb{F}^s \to \mathbb{F}^N$ by assigning values to the s edges of C and then considering the coefficient vector of the polynomial defined by the resulting circuit. The dimension of the image of Φ is at most s. There are only finitely many circuit topologies of size s so that the set of all polynomials computable in this manner is a finite union of sets with dimension at most s. For this union to be Zariski-dense in $\mathbb{F}^N$, we must have that $s \geq N$, proving the claim. □

A similar argument holds over fields of characteristic $p > 0$, but we will focus throughout on fields of characteristic zero. In algebraic complexity, we are interested in showing that certain *explicit* polynomials with certain properties are hard to compute. Thus, Proposition 2.6 is insufficient for our purposes since it does not construct any examples of hard polynomials.

Definition 2.7 (Algebraic Branching Programs): An *algebraic branching program* (*ABP*) over a field $\mathbb{F}$ is a directed acyclic graph A such that:

- The nodes V of A are partitioned into non-empty subsets $V = V_0 \sqcup \cdots \sqcup V_d$ for some integer $d \geq 0$ and we call V_i the i-th layer. Layer V_0 has exactly one node called the *source* s and layer V_d has exactly one node called the *sink* t.
- Each edge in A goes from layer V_{i-1} to V_i for some $i \in [d]$.
- Each edge e in A is labeled with a polynomial $\gamma_e \in \mathbb{F}[\overline{x}]$ of degree at most one.

For a path $\overline{e} = (e_1,\dots,e_d)$ from s to t, where each edge e_i goes from layer $i-1$ to layer i, define the polynomial $\gamma_{\overline{e}} := \prod_{i=1}^d \gamma_{e_i}$. Then A computes the polynomial $\sum_{\substack{\text{path } \overline{e} \\ \text{from } s \text{ to } t}} \gamma_{\overline{e}}$.

The *size* of an ABP A is the number of edges in A. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has $O\left(v^2\right)$ edges. The *width* of A is the maximum number of vertices in any layer of A. The *depth* of A is the number of layers.

Remark 2.8: Definition 2.7 is really the definition of a *layered* ABP, where each edge only goes from one layer to the next without skipping any layers. However, one can convert a non-layered ABP to a layered ABP with only a quadratic blowup in size.

Algebraic branching programs capture the complexity of computing a determinant.

Definition 2.9 (Determinantal Complexity): For a polynomial $f(\overline{x})$ over a field $\mathbb{F}$, the *determinantal complexity* of $f(\overline{x})$, denoted $\mathrm{dc}(f)$, is the smallest m such that there exists an $m \times m$ matrix M whose entries are degree ≤ 1 polynomials in $\mathbb{F}[\overline{x}]$ such that $\det_m(M) = f(\overline{x})$.

It is not clear that the determinantal complexity of a polynomial is always finite. We can easily compute monomials or even products of degree ≤ 1 terms as the determinant of a diagonal matrix. But it is not clear that if M_f and M_g are such that $\det\big(M_f\big) = f(\overline{x})$ and $\det\big(M_g\big) = g(\overline{x})$ how one constructs a matrix M_{f+g} such that $\det\big(M_{f+g}\big) = f(\overline{x}) + g(\overline{x})$. However, every polynomial can be computed by an algebraic branching program and the following lemma then gives that the determinantal complexity is always finite.

Lemma 2.10 ([Val79; Bür24]): Suppose that $f(\overline{x})$ can be computed by an algebraic branching program of size m. Then $\mathrm{dc}(f) \leq m$.

In fact, we can say much more about the connection between the complexity of branching programs and determinants:

Lemma 2.11 ([AF22, Lemma 3.6], [Val79, Theorem 1]): Let $g(\overline{y}) \in \mathbb{F}[\overline{y}]$ and suppose g can be computed by an algebraic branching program on r vertices. Then there is an $r \times r$ matrix A over $\mathbb{F}[\overline{y}]$ whose entries are polynomials in $\overline{y}$ of degree at most 1 such that

- $\det_r(A) = 1 + g(\overline{y})$, and
- for all $1 \leq k < r$, $\det_k(A_{[k],[k]}) = 1$.

Algebraic branching programs are closed under homogenization, as stated in the following lemma.

Lemma 2.12 ([AF22, Lemma 3.7]): Let $g(\overline{y}) \in \mathbb{F}[\overline{y}]$ and suppose g can be computed by an algebraic branching program on m vertices. Let z be a new indeterminate. Then there is a homogeneous polynomial $\widehat{g}(\overline{y}, z)$ such that $g = \widehat{g}(\overline{y}, 1)$ and $\widehat{g}$ can be computed by an algebraic branching program on m vertices.

2.2. Complexity Classes.

Definition 2.13 (VP): The class VP is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic circuit of size $\mathrm{poly}(n)$.

Definition 2.14 (VF): The class VF is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic formula of size $\mathrm{poly}(n)$.

Definition 2.15 (VBP): The class VBP is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic branching program of size $\mathrm{poly}(n)$.

Definition 2.16 (VNP): The class VNP is the set of polynomials $f(x_1, \ldots, x_n)$ such that there is some $g(x_1, \ldots, x_n, y_1, \ldots, y_m)$ in VP, where $m = \mathrm{poly}(n)$, such that

$$f(x_1, \ldots, x_n) = \sum_{\overline{a} \in \{0,1\}^m} g(x_1, \ldots, x_n, a_1, \ldots, a_m).$$

Valient's Criterion gives a useful characterization for what it means for $f(\overline{x})$ to be in VNP.

Theorem 2.17 ([Val79]): Let $f = \sum_{\overline{e}} f_{\overline{e}} x_1^{e_1} \cdots x_n^{e_n}$. Consider a function ϕ which take as input $\overline{e}$ and computes the coefficient $f_{\overline{e}}$. If ϕ is computable in polynomial time*, then $f \in \mathsf{VNP}$.

We give some discussion on the restriction of poly(n) degree for VP:

Remark 2.18: Note that via repeated squaring, one can compute polynomials of exponential degree with small circuits, even with restrictions on the fan-in of gates. However, most "interesting" polynomials, such as the determinant and permanent, fall into a "low-degree" regime where the degree of the polynomial is not significantly greater than the number of variables. Originally, Valiant was studying formulas, not circuits. One can easily see from Definition 2.2 that a formula of depth Δ must compute a polynomial of degree $O(\Delta)$ if we restrict to constant fan-in. See [Sap21] and [Gro] for more discussion.

2.3. **Basic Structural Results.** Notice that in our definition of an algebraic circuit,Definition 2.1, we don't allow division. However, certain polynomials are efficiently expressed as the quotient of two polynomials, such as $f(\overline{x}) = \frac{x^{d+1}-1}{x-1}$. It turns out that this is not an issue, as one can eliminate division gates without much blowup.

Lemma 2.19 ([Str73; HY11]): Suppose $f(x_1, \ldots, x_n)$ is a degree d polynomial computed by a circuit of size s and depth Δ with division gates. Then $f(x_1, \ldots, x_n)$ can be computed by a circuit of size $O\left(d^2 s^3\right)$ and depth $O(\log(s) \cdot \Delta)$.

Since we are often concerned solely with superpolynomial lower bounds, a polynomial blowup as in Lemma 2.19 is not of great concern.

Remark 2.20: Note that such division elimination works for formulas as well as algebraic branching programs. However, there are simple computational models where eliminating divisions efficiently is not possible. Recall that the *sparse* representation of a polynomial is when the complexity of a polynomial is given by the number of monomials. Consider the polynomial $x^d - 1$. Then $x^d - 1 = (x-1)(x^{d-1} + x^{d-2} + \cdots + x + 1)$ gives that while $x^d - 1$ has constant sparsity 2, it has a factor of sparsity d. Thus, the complexity of this model is not closed under divisions.

Clearly every formula is also a circuit so that the inclusion $\mathsf{VF} \subseteq \mathsf{VP}$ is immediate. Furthermore, it is clear that $\mathsf{VP} \subseteq \mathsf{VNP}$. What can be said about VBP as compared to VF and VP? The following can be proven by induction on the structure of a formula.

Lemma 2.21: Suppose that $f(\overline{x})$ is computed by a formula of size s. Then $f(\overline{x})$ can be computed by an algebraic branching program of size $O(s)$.

*Formally, we mean that ϕ is in the complexity class P/poly. However, saying "polynomial time" is sufficient for this purpose.

Lemma 2.22: Suppose that $f(\overline{x})$ is computed by an algebraic branching program of size s. Then $f(\overline{x})$ is computed by an algebraic circuit of size poly(s).

This follows from the fact that branching programs are computed by determinants (Lemma 2.10) and that the determinant has polynomial sized circuits:

Proposition 2.23: The $n \times n$ determinant has circuits of size poly(n).

Proof: *Sketch:* Compute Gaussian elimination generically on an $n \times n$ matrix of variables. This gives a poly(n) sized circuit with division for the determinant, as the determinant of a diagonal matrix is easy to compute and we can collapse all the elementary transformations to a single linear transformation. Afterwards, eliminate divisions with Lemma 2.19 to get a poly(n) sized algebraic circuit computing $\det_n$. □

With the above discussion, we get the following chain of inclusions.

Proposition 2.24:

$$\mathsf{VF} \subseteq \mathsf{VBP} \subseteq \mathsf{VP} \subseteq VNP$$

No strict separations between any of these classes is known. Showing a separation between any pair of these four classes is one of the biggest open problems in algebraic complexity.

2.3.1. *Oracle Complexity.* The notion of determinantal complexity (Definition 2.9) can be rephrased as asking how hard it is to compute a given polynomial assuming that computing the determinant is "free."We can generalize this notion an ask how hard it is to compute some polynomial $f(\overline{x})$ assuming that another polynomial $g(\overline{x})$ can be computed for free. For example, we will study how hard it is to compute a generator of an ideal I assuming that some fixed $0 \neq f(\overline{x}) \in I$ can be computed for free. We formalize this notion as follows:

Definition 2.25 (Oracle Circuits and Formulas): For a fixed polynomial $f(\overline{x}) \in \mathbb{F}[\overline{x}]$, an f*-oracle circuit* is an algebraic circuit defined in the same manner as Definition 2.1, however instead of just Σ and Π gates we now additionally allow $f(\overline{x})$ gates. Note that such gates may have some inputs set as wires and some inputs set as fixed constants from $\mathbb{F}$. We also define an f*-oracle formula* as an f-oracle circuit whose underlying graph is a tree.

2.3.2. *Computing Coefficients.*

Definition 2.26: For $f \in \mathbb{F}[\overline{x}, t]$ and an integer $i \in \mathbb{Z}_{\geq 0}$, denote by $\mathrm{coeff}_{t^i}(f)$ the coefficient of t^i in f when f is viewed as a univariate polynomial in t over the ring $\mathbb{F}[\overline{x}]$.

Lemma 2.27 ([DG25, Lemma 2.29]): Let $g \in \mathbb{F}[\overline{x}, t]$, and let $f \in \mathbb{F}[\overline{x}, t]$ be a degree-d polynomial such that for each $\alpha \in \mathbb{F}$, the polynomial $f(\overline{x}, \alpha)$ is computed by a g-oracle circuit C_α of size at most

s and depth at most Δ. Assume that $|\mathbb{F}| \geq d+1$. Then, for every $0 \leq i \leq d$, there exists a g-oracle circuit D_i of size $O(ds)$ and depth $\Delta+1$ that computes $\mathrm{coeff}_{t^i}(f)$, with its top gate being an addition gate and bottom Δ layers consisting of $d+1$ copies of g-oracle circuits of the form C_α. Moreover, if the top gate of each C_α is an addition gate, then the depth bound can be improved to Δ.

Proof: By definition,

$$f = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(f) t^i.$$

Let $\alpha_1, \ldots, \alpha_{d+1}$ be $d+1$ distinct elements in $\mathbb{F}$. Then we have that

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

The $(d+1) \times (d+1)$ matrix $(\alpha_i^{d+1-j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ is a Vandermonde matrix. It's determinant is $\prod_{i<j}(\alpha_j - \alpha_j)$. As all the α_i are distinct, this determinant is nonzero so that the matrix is invertible. Thus, there exists a $(d+1) \times (d+1)$ matrix $(z_{i,j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ such that

$$\begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,d} & z_{1,d+1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,d} & z_{2,d+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{d,1} & z_{d,2} & \cdots & z_{d,d} & z_{d,d+1} \\ z_{d+1,1} & z_{d+1,2} & \cdots & z_{d+1,d} & z_{d+1,d+1} \end{pmatrix} \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

Fix $0 \leq i \leq d$. We now have an explicit expression for $\mathrm{coeff}_{t^i}(f)$:

$$\mathrm{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\overline{x}, \alpha_j) \tag{1}$$

With Equation (1), we can now describe the circuit for $\mathrm{coeff}_{t^i}(f)$. The scalars $\alpha_1, \ldots, \alpha_{d+1}$ are fixed a priori, and hence so are the $z_{i,j}$ for $i, j \in [d+1]$. Recall that scalar multiplications along wires are free. For each $j \in [d+1]$, computing $f(\overline{x}, \alpha_j)$ requires a single g-oracle circuit C_{α_j} of size at most s and depth Δ. Hence, evaluating all $d+1$ such terms requires total size $O(ds)$. The top-level summation uses one addition gate and $d+1$ scalar multiplications by the $z_{d+1-i,j}$ along $d+1$ wires. Therefore, Equation (1) yields a g-oracle circuit D_i of size $O(ds+d) = O(ds)$ and depth $\Delta+1$ computing $\mathrm{coeff}_{t^i}(f)$. Moreover, if the top gate of each C_{α_j} is an addition gate, then the top two layers of D_i consist entirely of additions and can be merged to reduce the depth bound to Δ. $\square$

Remark 2.28: The condition $|\mathbb{F}| \geq d+1$ in Lemma 2.27 is required for interpolation. We note that if $f(\overline{x})$ is given not as a black box but as a circuit, then its homogeneous components can be extracted in a gate-by-gate fashion [Str73], which requires no lower bound on $|\mathbb{F}|$.

In [DG25], we give a new tool for interpolation in the setting of algebraic complexity. The *Isolation Lemma* of Mulmuley, Vazirani, and Vazirani [MVV87] has seen much application in combinatorics and complexity theory. The idea is that by adding random restrictions to a solution space, one can efficiently reduce the number of solutions from many to a single solution which can now be found in a smaller search space. We use the formulation of Klivans and Spielman [KS01] which applied the Isolation Lemma to the context of polynomial identity testing of sparse polynomials.

Lemma 2.29 ([KS01, Lemma 4]): Let C be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$. Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \geq K\ell/\varepsilon$. Then, with probability at least $1-\varepsilon$, there is a unique form in C of minimum value at $z_1, \dots, z_\ell$.

A simple rephrasing of Lemma 2.29 allows us to isolate a term of a multivariate polynomial $f(y_1, \dots, y_\ell)$ under a random monomial substitution $y_i \mapsto w^{z_i}$:

Corollary 2.30 ([DG25, Corollary 2.27]): Let $K \in \mathbb{Z}_{\geq 0}$. Let A be any ring, and let $f(y_1, \dots, y_\ell) \in A[\overline{y}]$ be a polynomial whose individual degree in each variable is at most K, i.e., $\deg_{y_i}(f) \leq K$ for all $i \in [\ell]$. Fix $\varepsilon > 0$. Choose $z_1, \dots, z_\ell$ independently and uniformly at random from $\{0, \dots, M\}$, where $M \geq K\ell/\varepsilon$. Let w be a new indeterminate, and define $\phi : A[y_1, \dots, y_\ell] \to A[w]$ to be the ring homomorphism sending $y_i \mapsto w^{z_i}$ for each $i \in [\ell]$. Then, with probability at least $1-\varepsilon$, there exists a unique monomial $\mathfrak{m} = y_1^{e_1} \cdots y_\ell^{e_\ell}$ among all monomials of f such that $\phi(\mathfrak{m}) = w^{e_1 z_1 + \cdots + e_\ell z_\ell}$ attains the minimum degree in w.

The advantage of Corollary 2.30 is when we have a multivariate polynomial in which there are multiple terms of the same total degree. In this situation, suppose that we want a circuit for a single term. Then Lemma 2.27 is insufficient for this purpose as it separates terms by the degree in a single variable. Dealing with multiple variables by repeatedly applying Lemma 2.27 would yield an exponential blowup in the total circuit size. A similar idea would be to apply a *Kronecker* substitution:

Lemma 2.31 ([Kro82]): Write $g(\overline{x}) = \sum_{\overline{e}} \overline{x}^{\overline{e}}$. Suppose that $\deg(g) < d$. Then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

One could apply the Kronecker map to our multivariate polynomial and then apply Lemma 2.27 since all terms would have distinct total degree. However, the total degree in w, the new variable, would be exponential so that the resulting circuit would have large size. Here, Corollary 2.30 argues there is an alternative assignment of weights maps $x_i \mapsto w^{z_i}$ rather than $x_i \mapsto w^{d^i}$ such that there is a unique term of lowest total degree in w. Thus, if we can argue that *any* of our terms of the same total degree suffice, we can apply Corollary 2.30 followed by Lemma 2.27 to get a small circuit computing a single term. To the best of our knowledge, this is the first usage of the Isolation Lemma of [MVV87; KS01] in algebraic complexity for the purposes of isolating terms in a polynomial.

2.3.3. *Constant Depth Circuits.* Much recent work in algebraic complexity revolves around the constant-depth regime. There is good reason for this, as it has been shown that to show circuit size lower bounds on general arithmetic circuits, it suffices to show lower bounds on circuits of low depth. First, it was shown that it suffices to prove lower bounds for depth-4 circuits:

Theorem 2.32 ([AV08; Koi12]): Suppose $f(x_1,\ldots,x_n) \in \mathsf{VP}$ is of degree d. Then $f(x_1,\ldots,x_n)$ can be computed by a $\Sigma\Pi\Sigma\Pi$ circuit of size $2^{O\left(\sqrt{d}\cdot\log^2 d\right)}$.

This work was then improved on to show that one only needs to provide lower bounds for depth-3 circuits:

Theorem 2.33 ([Gup+13]): Suppose $f(x_1,\ldots,x_n) \in \mathsf{VP}$ is of degree d and computable by a circuit of size s. Then $f(x_1,\ldots,x_n)$ can be computed by a $\Sigma\Pi\Sigma$ circuit of size $2^{O\left(\sqrt{d\log(d)\log(n)\log(s)}\right)}$.

However, it is hard to obtain lower bounds in the constant depth regime. We have only very recently obtained lower bounds of constant depth circuit size for explicit polynomials.

Definition 2.34 (Iterated Matrix Multiplication): Let $n, d \in \mathbb{Z}_{>0}$ be parameters. Let $X_1, X_2, \ldots, X_d$ be d independent matrices of variables. Then the *Iterated Matrix Multiplication polynomial*, $\mathrm{IMM}_{n,d}$, is the polynomial in position $(1,1)$ of the matrix product $X_1 \cdots X_d$.

Theorem 2.35 ([LST25; For24]): Let $d = o(\log n)$. Then for any $\Delta \geq 1$, computing $\mathrm{IMM}_{n,d}$ with a depth Δ circuit requires that circuit to have size at least

$$n^{\Theta\left(\frac{1}{F_{\Delta+2}-1}/\Delta\right)}$$

where F_k is the k-th Fibonacci number.

2.4. **Border Complexity.**

Definition 2.36 (Border Computation): We say that $f(\overline{x}) \in \mathbb{F}[x_1,\ldots,x_n]$ is *border computed* by a circuit C over $\mathbb{F}(\varepsilon)$ if C computes some polynomial $h \in \mathbb{F}(\varepsilon)[x_1,\ldots,x_n]$, and $h - f \in \varepsilon \cdot \mathbb{F}[[\varepsilon]][x_1,\ldots,x_n]$, where $\mathbb{F}[[\varepsilon]]$ denotes the ring of formal power series in ε. We will denote by $\overline{\mathrm{size}}(f)$ the size of the smallest circuit over $\mathbb{F}(\varepsilon)$ border computing $f(\overline{x})$.

In the settings where $\mathbb{F}[\varepsilon][x_1,\ldots,x_n]$ can be equipped with the Euclidean topology in a natural way, such as $\mathbb{F}=\mathbb{C}$ or $\mathbb{R}$, this would imply that $h \to g$ as $\varepsilon \mapsto 0$. However, this does not mean that we could just substitute ε by zero in C and obtain a circuit exactly computing g since C may use scalars in $\mathbb{F}(\varepsilon)$, such as $1/\varepsilon$, that are not well-defined at $\varepsilon=0$.

Example 2.37 ([DL25]): The *Waring Rank* of a degree d homogeneous polynomial $f(\overline{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\overline{x}),\ldots,\ell_r(\overline{x})$ such that

$$f(\overline{x})=\ell_1(\overline{x})^d+\cdots+\ell_r(\overline{x})^d.$$

We denote this minimum r by $\mathrm{WR}(f)$. Note that $\mathrm{WR}(x^{d-1}y)=d$ [Old40]. However, we have that

$$x^{d-1}y=\lim_{\varepsilon\to 0}\frac{1}{d\varepsilon}((x+\varepsilon y)^d-x^d).$$

Here, we say that the *Border Waring Rank*, $\overline{\mathrm{WR}}(f)$, is equal to 2. This is an example where we cannot just set $\varepsilon=0$ and obtain a more efficient representation in our computational model.

For any of the complexity classes we have defined above, we can define a border analogue. This corresponds to a Zariski closure of the complexity class when the class of polynomials is viewed as a subset of affine space via their coefficient vectors. We give the following definition for the closure of VP, but defining the closures of VF, VBP, and VNP is completely analogous.

Definition 2.38 ($\overline{\mathsf{VP}}$): The class $\overline{\mathsf{VP}}$ is the set of polynomials $f(x_1,\ldots,x_n)$ over $\mathbb{F}$ of degree $\mathrm{poly}(n)$ which are border computable by an algebraic circuit of size $\mathrm{poly}(n)$ over $\mathbb{F}(\varepsilon)$.

An important question in algebraic complexity is asking what the power of border computation is. Often, this question takes the form of studying the inclusion $\mathcal{C}\subseteq\overline{\mathcal{C}}$ for some complexity class $\mathcal{C}$ and finding a description of $\overline{\mathcal{C}}\setminus\mathcal{C}$. Questions of this form have already attracted much study in algebraic complexity.

Example 2.39:

- Debordering depth-2 circuits is simple and we recover that $\overline{\Sigma\Pi}=\Sigma\Pi$ and $\overline{\Pi\Sigma}=\Pi\Sigma$ [DL25].
- Debordering results for depth-3 and depth-4 circuits already would have strong implications towards polynomial identity testing [AV08; Koi12; Gup+13].
- Early work in classical algebraic geometry discusses debordering Waring Rank [Pal06; Ter11].
- Forbes conjectured that if $f\in\overline{\Sigma^s\wedge^d\Sigma}$ then $f\in\Sigma^{s'}\wedge^d\Sigma$ for $s'\in\mathrm{poly}(s,n,d)$ [Shp16].

Another setting of border computation is asking if certain transformations and operations are more efficient in the border setting. For example, we know that to interpolate a degree d polynomial, you must evaluate your polynomial at $d+1$ distinct points. However in the border setting, we can actually border compute the coefficient $\mathrm{coeff}_{t^r}(f(t))$ using only $r+1$ points.

Example 2.40 ([DL25]): In order to interpolate the coefficients of $f(\overline{x},t)=\sum_{i=0}^{d}\mathrm{coeff}_{t^i}(f)\cdot t^i$ over $\mathbb{F}$ as in Lemma 2.27, one needs that $|\mathbb{F}|\geq d+1$. However, we can border compute $\mathrm{coeff}_{t^r}(f)$ if $|\mathbb{F}|\geq r+1$.

Suppose $\mathbb{F}$ has distinct elements $\{\alpha_0,\dots,\alpha_r\}$. Then, there exists elements $\beta_0,\dots,\beta_r\in\mathbb{F}$ such that

$$\mathrm{coeff}_{t^r}(f)=\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r}\beta_i\cdot f(\overline{x},\varepsilon\alpha_i)\right).$$

Indeed let $g(\overline{x},t)=\sum_{i=0}^{r}\mathrm{coeff}_{t^i}(f)\cdot t^i$ and $h(\overline{x},t)=f(\overline{c},t)-g(\overline{x},t)$. Then standard interpolation on $\varepsilon\alpha_i$ gives us that there exists field constants $\beta_0,\dots,\beta_r$ such that $\varepsilon^r\,\mathrm{coeff}_{t^r}(f)=\sum_{i=0}^{r}\beta_i\cdot g(\varepsilon\alpha_i)$. Thus, we have that

$$\begin{aligned}
&\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r}\beta_i\cdot f(\overline{x},\varepsilon\alpha_i)\right)\\
=&\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r}\beta_i\cdot g(\overline{x},\varepsilon\alpha_i)\right)+\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r}\beta_i\cdot h(\overline{x},\varepsilon\alpha_i)\right)\\
=&\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^r}\left(\sum_{i=0}^{r}\beta_i\cdot g(\overline{x},\varepsilon\alpha_i)\right)+0=\mathrm{coeff}_{t^r}(f).
\end{aligned}$$

3. Preliminaries on Straightening Laws

3.1. **Determinantal Ideals.** Let $\mathbb{F}$ be a field. Let $X=(x_{i,j})_{\substack{1\leq i\leq n\\ 1\leq j\leq m}}$ be a $n\times m$ matrix of variables. We will denote by $\mathbb{F}[X]$ the polynomial ring $\mathbb{F}[x_{i,j}\mid 1\leq i\leq n,\ 1\leq j\leq m]$. We can natural associate a product of minors of X to a pair of Young tableau where the minors are described by the entries of the Young tableau.

Definition 3.1 (Bitableau, bideterminant): Let $\sigma=(\sigma_1\geq\cdots\geq\sigma_k)$ be a partition. An (n,m)-*bitableau* $(S\mid T)$ of shape σ is a pair of two Young tableau of shape σ with where the entries of S come from $[n]$ and the entries of T come from $[m]$. When n and m are clear from context, we just call $(S\mid T)$ a *bitableau*. For each $i\in[\widehat{\sigma}_1]$, write the i-th row of S and the i-th row of T as $S_i=(S(i,1),\dots,S(i,\sigma_i))$ and $T_i=(T(i,1),\dots,T(i,\sigma_i))$ respectively. The $\sigma_i\times\sigma_i$ minor of X defined by S_i and T_i is the matrix

$$\left(x_{S(i,a),T(i,b)}\right)_{1\leq a,b\leq\sigma_i}=\begin{pmatrix} x_{S(i,1),T(i,1)} & x_{S(i,1),T(i,2)} & \cdots & x_{S(i,1),T(i,\sigma_i)}\\ x_{S(i,2),T(i,1)} & x_{S(i,2),T(i,2)} & \cdots & x_{S(i,2),T(i,\sigma_i)}\\ \vdots & \vdots & \ddots & \vdots\\ x_{S(i,\sigma_i),T(i,1)} & x_{S(i,\sigma_i),T(i,2)} & \cdots & x_{S(i,\sigma_i),T(i,\sigma_i)}\end{pmatrix}.$$

Then the associated *bideterminant* $(S \mid T)(X)$ is the product of all such minors over $i \in [\widehat{\sigma}_1]$:

$$(S \mid T)(X) := \prod_{i=1}^{\widehat{\sigma}_1} \det_{\sigma_i}\left(x_{S(i,a),T(i,b)}\right)_{1\le a,b\le \sigma_i}.$$

An $n \times n$ determinant is homogeneous of degree n so that $(S \mid T)(X)$ is a homogeneous polynomial of degree $|\sigma|$.

Example 3.2: If $\sigma = (4,2,1)$ and $(S \mid T)$ is the bitableau

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 \\ \cline{1-2} 4 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 \\ \cline{1-2} 3 \\ \cline{1-1} \end{array} \right),$$

then the bideterminant $(S \mid T)(X)$ is given by

$$\begin{vmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{vmatrix} \begin{vmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{vmatrix} \begin{vmatrix} x_{4,3} \end{vmatrix}.$$

The following lemma shows that every monomial in $\mathbb{F}[X]$ can be expressed as a bideterminant, and hence the bideterminants span $\mathbb{F}[X]$ as an $\mathbb{F}$-vector space. The proof is immediate from the definition.

Lemma 3.3: A degree d monomial $\prod_{i=1}^{d} x_{r_i,c_i}$ is given by a bideterminant:

$$\left(\begin{array}{|c|} \hline r_1 \\ \hline r_2 \\ \hline \vdots \\ \hline r_d \\ \hline \end{array} \;\middle|\; \begin{array}{|c|} \hline c_1 \\ \hline c_2 \\ \hline \vdots \\ \hline c_d \\ \hline \end{array} \right)(X) = \prod_{i=1}^{d} x_{r_i,c_i}.$$

Thus, the bideterminants span $\mathbb{F}[X]$.

While the bideterminants span $\mathbb{F}[X]$, there is a specific subset of bideterminants, the *standard* bideterminants, which form an $\mathbb{F}$-basis of $\mathbb{F}[X]$.

Definition 3.4 (Standard bitableau, standard bideterminant): We call a bitableau $(S \mid T)$ and its associated bideterminant $(S \mid T)(X)$ *standard* if S and T are both conjugate semistandard Young tableaux.

The following theorem is known in the literature as the *straightening law*.

Theorem 3.5 ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3]): Let $(S \mid T)(X)$ be a bideterminant of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X),$$

where $(A \mid B)(X)$ is a standard bideterminant of shape $\tau \geq_{\text{lex}} \sigma$, and $c_{A,B} \in \mathbb{Z}$ when $\text{char}(\mathbb{F}) = 0$, while $c_{A,B} \in \mathbb{Z}/p\mathbb{Z}$ when $\text{char}(\mathbb{F}) = p > 0$.

The fact that bideterminants are homogeneous polynomials, together with Lemma 3.3 and Theorems 3.5, imply the following corollary.

Corollary 3.6: The standard bideterminants form an $\mathbb{F}$-basis of $\mathbb{F}[X]$. In particular, since bideterminants are homogeneous with respect to total degree, the degree-d component of $\mathbb{F}[X]$ has as basis the standard bideterminants of shape σ such that $|\sigma| = d$.

Let $I^{\text{det}}_{n,m,r}$ be the ideal of $\mathbb{F}[X]$ generated by the $r \times r$ minors of X. Note that this ideal is a $\mathbb{F}$-subspace of $\mathbb{F}[X]$. We have the following crucial statement on the $\mathbb{F}$-basis of this ideal.

Proposition 3.7 ([BC03, Corollary 1.7]): Polynomials $f \in I^{\text{det}}_{n,m,r}$ are supported by standard bideterminants of shape σ such that $\sigma_1 \geq r$; that is, by those whose first row has length at least r.

The notion of the total degree of a polynomial is too coarse for differentiating terms in the expansion of a polynomial $f \in I^{\text{det}}_{n,m,r}$ with respect to the standard basis. Thus, we describe a grading of the polynomial ring $\mathbb{F}[X]$ that is better suited for this purpose.

Definition 3.8: The polynomial ring $\mathbb{F}[X]$ has a natural $(\mathbb{N}^n \oplus \mathbb{N}^m)$-grading. Let $\overline{e}_i \in \mathbb{N}^n$ be the standard basis vector with a 1 in position i and 0's in the other positions. Define $\overline{e}_i \in \mathbb{N}^m$ the same way. Then we assign $x_{i,j}$ to have degree $\overline{e}_i \oplus \overline{e}_j$. We call this assignment the *multidegree*. We say a polynomial $f \in \mathbb{F}[X]$ is *multihomogeneous* of multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_m\overline{e}_m)$ if every monomial in f has multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_m\overline{e}_m)$.

The following lemma gives the multidegree of bideterminants. In particular, the multidegree of canonical and anti-canonical bideterminants will allow us to extract terms of a particular shape.

Lemma 3.9: Fix a natural number $n \in \mathbb{Z}_{\geq 0}$. Let $\sigma = (\sigma_1 \geq \cdots \geq \sigma_d)$ be a partition such that the length σ_i of each row is at most n. Let S, T be two conjugate semistandard Young tableaux. Then a bideterminant $(S \mid T)(X)$ is multihomogeneous of multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_n\overline{e}_n)$, where s_i is equal to the number of i's that appear in S and t_j is equal to the number of j's that appear in T. In particular, $(K_\sigma \mid K_\sigma)(X)$ is multihomogeneous of multidegree $(\widehat{\sigma}_1\overline{e}_1 + \cdots + \widehat{\sigma}_n\overline{e}_1) \oplus (\widehat{\sigma}_1\overline{e}_1 + \cdots + \widehat{\sigma}_n\overline{e}_1)$. Moreover, for distinct partitions $\sigma \neq \tau$, the multidegrees of $(K_\sigma \mid K_\sigma)(X)$ and $(K_\tau \mid K_\tau)(X)$ are distinct, and the same holds for the multidegrees of $(\overline{K}_\sigma \mid \overline{K}_\sigma)(X)$ and $(\overline{K}_\tau \mid \overline{K}_\tau)(X)$.

Proof: The multidegree of a standard bideterminant $(S \mid T)(X)$ is immediate from Definition 3.1. To see that for distinct partitions $\sigma \neq \tau$ the multidegrees of $(K_\sigma \mid K_\sigma)(X)$ and $(K_\tau \mid K_\tau)(X)$ are distinct, we make the following observation. The number of 1's in K_σ is the coefficient of $\overline{e}_1$ in the multidegree, the number of 2's in K_σ is the coefficient of $\overline{e}_2$, and so on. This allows us to get the values $\widehat{\sigma}_1,\dots,\widehat{\sigma}_n$ which in turn completely determines σ. $\square$

3.1.1. *Determinantal Varieties.*

Definition 3.10 (Determinantal Variety): Let $\mathbb{F}$ be an algebraically closed field. Let $V_{n,m,r} \subseteq \mathbb{A}^{nm}_{\mathbb{F}}$ be the set of $n\times m$ matrices of rank at most r. This is the vanishing locus of all $(r+1)\times(r+1)$ minors of $X=(x_{i,j})_{\substack{1\le i\le n\\ 1\le j\le m}}$.

Proposition 3.11: The variety $V_{n,m,r}$ is irreducible.

Proof: Let $\widetilde{V}_{n,m,r} = \{(M,W)\in M_{n\times m}(\mathbb{F})\times \mathrm{Gr}(n,n-r) \mid W\subseteq \ker(M)\}$. Consider the projection map $\pi_2\colon \widetilde{V}_{n,m,r}\to \mathrm{Gr}(n,n-r)$. Then, recall that $\mathrm{Gr}(n,n-r)$ is irreducible (Proposition 1.5). Let $W\in \mathrm{Gr}(n,n-r)$. The fiber over W of π_2 is the set of all $n\times m$ matrices M which vanish on W. This fiber is isomorphic to $\mathbb{A}^{m\cdot r}_{\mathbb{F}}$. Thus, since π_2 surjects onto an irreducible variety and has irreducible fibers all of the same dimension, we must have that $\widetilde{V}_{n,m,r}$ is irreducible [Har92, Theorem 11.14].

Consider $\pi_1\colon \widetilde{V}_{n,m,r}\to V_{n,m,r}$ given by the other projection map. This map is surjective since $\pi_1^{-1}(M)=(M,\ker M)$. Thus, since $\widetilde{V}_{n,m,r}$ is irreducible, we get that $V_{n,m,r}$ is irreducible. $\square$

Proposition 3.12: The variety $V_{n,m,r}$ has dimension $(n-r)r+mr$ and thus codimension $(n-r)(m-r)$ in $\mathbb{A}^{nm}_{\mathbb{F}}$.

Proof: Consider the map π_1 from the prior proof. In addition to it being surjective, we have that it is generically injective. Specifically, consider the Zariski-dense subset of matrices which have rank equal to r. Then for each of these matrices M, their kernel $\ker M$ has dimension exactly $n-r$ by Rank-Nullity. Thus, the preimage of M must be $(M,\ker M)$ exactly. Generically, this means that each fiber is zero-dimensional so that

$$\dim V_{n,m,r} = \dim \widetilde{V}_{n,m,r} = \dim \mathrm{Gr}(n,n-r)+\dim \mathbb{A}^{m\cdot r}_{\mathbb{F}} = (n-r)r+mr.$$

$\square$

3.2. **Pfaffian Ideals.** We now introduce notation for Pfaffian ideals. Let $\mathbb{F}$ be a field. Let $x_{i,j}$ be indeterminates for $1\le i<j\le 2n$ and let $x_{j,i}:=-x_{i,j}$ for $1\le j<i\le 2n$ and $x_{i,i}=0$ for $1\le i\le 2n$. Thus, $X=(x_{i,j})_{1\le i,j\le 2n}$ is a skew-symmetric $2n\times 2n$ matrix of variables. For notational consistency with the prior subsections, we denote the polynomial ring $\mathbb{F}[x_{i,j}\mid 1\le i<j\le 2n]$ by $\mathbb{F}[X]$.

Recall that for a skew-symmetric matrix X, the determinant when viewed as a polynomial in the entries of the skew-symmetric matrix is a perfect square of some other polynomial. The square root of $\det_{2n}(X)$ is called the *Pfaffian* of X. For the Pfaffian of a $2n \times 2n$ matrix $M \in \mathbb{F}^{2n\times 2n}$, we use the notation pfaff_{2n}. The Pfaffian has a more intrinsic definition given as follows:

$$\mathrm{pfaff}_{2n}(X) = \frac{1}{2^n n!} \sum_{\sigma \in \mathfrak{S}_{2n}} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} x_{\sigma(2i-1),\sigma(2i)}. \tag{2}$$

Just like the determinant, this is a polynomial with coefficients ± 1 so that it is well defined over any field. To see this, note that every monomial in Equation (2) appears exactly $2^n n!$ times. We also have that if m is odd, then $\det_m(X) = 0$ when X is a $m \times m$ skew-symmetric matrix so that $\mathrm{pfaff}_m(X) = 0$. Thus, we only consider Pfaffians on matrices of even order.

Pfaffians are deeply connected to determinants, as indicated by the following lemma:

Lemma 3.13 ([AF22, Lemma 2.27]): Let X be a $2n \times 2n$ skew-symmetric matrix and let E be an arbitrary $2n \times 2n$ matrix. Then $EXE^\top$ is also skew-symmetric and $\mathrm{pfaff}_{2n}(EXE^\top) = \det_{2n}(E)\,\mathrm{pfaff}_{2n}(X)$.

Since the Pfaffian is only well-defined for skew-symmetric matrices, we cannot consider all possible row and column combinations when looking at Pfaffians defined on minors of a matrix.

Definition 3.14 (Principal bitableau, bipfaffian): A bitableau $(S \mid T)$ is *principal* if $S = T$, i.e., if the submatrix of X with rows given by S and columns given by T is principal. In this case, we denote $[T] := (T \mid T)$. Since $m \times m$ determinants of skew-symmetric matrices are zero for odd m, we will assume all partitions have only even row lengths in this case. In particular, $|\sigma|$ will always be divisible by 2.

Let σ be a partition such that the length of each row of σ is even, and let T be a Young tableau of shape σ. Writing the rows as $T_i = (T(i,1), \dots, T(i,\sigma_i))$, the submatrix of X with rows and columns given by T_i is skew-symmetric as well and thus has a well-defined *principal pfaffian*. The associated *bipfaffian*[†] $[T](X)$ is the product of all such Pfaffians defined by the rows of T:

$$[T](X) := \prod_{i=1}^{\widehat{\sigma}_1} \mathrm{pfaff}\left(x_{T(i,a),T(i,b)}\right)_{1 \le a,b \le \sigma_i}.$$

Note that the total degree $\deg([T](X))$ is equal to $|\sigma|/2$.

The following lemma shows that every monomial in $\mathbb{F}[X]$ can be expressed as a bipfaffian and hence the bipfaffians span $\mathbb{F}[X]$. The proof is immediate from the definition.

[†] [AF22] call this a *standard monomial*. We deviate from this as standard monomial is a more general term which also refers to standard bideterminants and other analogues in standard monomial theory.

Lemma 3.15: A degree d monomial $\prod_{i=1}^{d} x_{r_i,c_i}$ where for each $1 \le i \le d$, $r_i < c_i$ is given by a bipfaffian:

$$\left[\begin{array}{|c|c|} \hline r_1 & c_1 \\ \hline r_2 & c_2 \\ \hline \vdots & \vdots \\ \hline r_d & c_d \\ \hline \end{array}\right](X) = \prod_{i=1}^{d} x_{r_i,c_i}.$$

Analogous to the classical determinantal case, a specific subset of bipfaffians form an $\mathbb{F}$-basis of $\mathbb{F}[X]$.

Definition 3.16 (Standard bipfaffian): A bipfaffian $[S](X)$ is *standard* if S is a conjugate semistandard Young tableau.

As in the determinantal case, the Pfaffian case also admits a straightening law.

Theorem 3.17 ([dCP76, Theorem 6.5], [HT92, §5]): Let S be a Young tableau of shape σ. Then $[S](X)$ can be uniquely expressed as a linear combination

$$[S](X) = \sum_{A} c_A [A](X),$$

where $[A](X)$ is a standard bipfaffian of shape $\tau \ge_{\text{lex}} \sigma$, and $c_A \in \mathbb{Z}$ when $\text{char}(\mathbb{F}) = 0$, while $c_A \in \mathbb{Z}/p\mathbb{Z}$ when $\text{char}(\mathbb{F}) = p > 0$.

Lemma 3.15 and Theorems 3.17 together imply the following corollary.

Corollary 3.18: The standard bipfaffians form an $\mathbb{F}$-basis of $\mathbb{F}[X]$. In particular, since bipfaffians are homogeneous with respect to total degree, the degree-d component of $\mathbb{F}[X]$ has as basis the standard bipfaffians of shape σ such that $|\sigma| = 2d$. Note that if $|\sigma| = 2d$, then the number of rows $\widehat{\sigma}_1$ of σ is at most d as each non-empty row of σ has length at least 2.

Let $I_{2n,2r}^{\text{pfaff}}$ be the ideal of $\mathbb{F}[X]$ generated by the $2r \times 2r$ principal bipfaffians of X. We have the following crucial statement.

Proposition 3.19 ([HT92, §5]): Polynomials $f \in I_{2n,2r}^{\text{pfaff}}$ are supported by standard bipfaffians of shape σ such that $\sigma_1 \ge 2r$.

Next, we describe a grading of the polynomial ring $\mathbb{F}[X]$ that is useful for the proof.

Definition 3.20: The polynomial ring $\mathbb{F}[X]$ has a natural $\mathbb{N}^{2n}$-grading. Let $\overline{e}_i \in \mathbb{N}^{2n}$ be the standard basis vector with a 1 in position i and 0's in the other positions. Then we assign $x_{i,j}$ to have degree $\overline{e}_i + \overline{e}_j$. We call this assignment the *multidegree*. We say a polynomial $f \in \mathbb{F}[X]$ is *multihomogeneous* of multidegree $s_1\overline{e}_1 + \cdots + s_{2n}\overline{e_{2n}}$ if every monomial in f has multidegree $s_1\overline{e}_1 + \cdots + s_{2n}\overline{e_{2n}}$.

The following lemma gives the multidegree of bipfaffians. It follows immediately from the definition and is proven similarly to Lemma 3.9 and thus we omit the proof.

Lemma 3.21: Fix a natural number $n \in \mathbb{Z}_{\geq 0}$. Let $\sigma = (\sigma_1 \geq \cdots \geq \sigma_k)$ be a partition such that the length of each row is at most $2n$. Let S be a conjugate semistandard Young tableau. Then a bipfaffian $[S](X)$ is multihomogeneous of multidegree $s_1\bar{e}_1 + \cdots + s_{2n}\bar{e}_{2n}$, where s_i is equal to the number of i's that appear in S. In particular, $[K_\sigma](X)$ is multihomogeneous of multidegree $\widehat{\sigma}_1\bar{e}_1 + \cdots + \widehat{\sigma}_{2n}\bar{e}_{2n}$. Moreover, for distinct partitions $\sigma \neq \tau$, the multidegrees of $[K_\sigma](X)$ and $[K_\tau](X)$ are distinct, and the same holds for the multidegrees of $[\overline{K}_\sigma](X)$ and $[\overline{K}_\tau](X)$.

4. Prior Work

4.1. **Generalities on Complexity in Ideals.** Roughly speaking, the complexity of an ideal I under a given algebraic model (such as algebraic circuits or algebraic branching programs) is defined as the minimum complexity, in that model, of any nonzero polynomial $f \in I$ [Gro20]. Proving lower bounds for ideals is harder than for individual polynomials, since one must establish a bound for every nonzero $f \in I$ rather than just a single polynomial. This motivates the search for a property of the following form: if an arbitrary nonzero $f \in I$ is efficiently computable in a given model, then some polynomial from a distinguished set $S = \{g_1, \dots, g_k\}$ (often a natural generating set of I or some related set with desirable properties) is also efficiently computable in the given model. In this way, proving lower bounds for arbitrary nonzero $f \in I$ reduces to proving them for $g_1, \dots, g_k$. We call such a property a *closure result*, by analogy with closure under factorization in the principal ideal case. In short, closure results reduce the task of showing hardness for all polynomials in I to proving hardness for a small set of distinguished polynomials.

The general question we want to study is of the following form:

Question 4.1: Say that $I = \langle g_1, \dots, g_k \rangle$. Let $0 \neq f \in I$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$.

4.1.1. *Principal Ideals.* Let us consider the special case of Question 4.1 where $I = \langle g \rangle$ is a *principal ideal*. Then our question is now the following:

Question 4.2: Let $0 \neq f \in \langle g \rangle$. Does g have a small f-oracle circuit?

The affirmative was actually conjectured by Bürgisser:

Conjecture 4.3 ([Bür00, Conjecture 8.3]): If g is a factor of $f \neq 0$, then $\mathrm{size}(g) \leq \mathrm{poly}(\mathrm{size}(f), \deg(g))$.

There has been much progress on Conjecture 4.3. Factoring in algebraic complexity is well studied, see the survey of Bhargav, Dwivedi, and Saxena for an overview of recent progress [BDS26]. For example, it is known that for fields of characteristic zero that $\mathrm{size}(g) \leq \mathrm{poly}(\mathrm{size}(f), \deg(f))$ [Kal87; Bür00]. Compare the use of $\deg(f)$ here to $\deg(g)$ in Conjecture 4.3. Thus, the most interesting regime to study here is the situation where the degree of the factor g is significantly smaller than the degree of the whole polynomial f. This result is only known in the setting of border computation over fields of characteristic zero:

Theorem 4.4 ([Bür04, Theorem 1.3]): Let $\mathbb{F}$ be a field of characteristic zero. Let $f(\overline{x}) \neq 0$ be an n-variate polynomial. If $g(\overline{x})$ is a factor of $f(\overline{x})$, then $\overline{\text{size}}(g) \leq \text{poly}(\text{size}(f), \deg(g), n)$.

Bürgisser's result can actually be formulated in the setting of oracle complexity, so that we get a $\text{poly}(\deg(g), n)$ size f-oracle circuit which approximately computes $g(\overline{x})$.

4.1.2. *Factoring versus Polynomial Identity Testing.*

Definition 4.5 (Polynomial Identity Testing (PIT)): Given black box access to an n-variate, degree d polynomial $f(x_1, \ldots, x_n) \in \mathbb{F}[x_1, \ldots, x_n]$, determine if $f(x_1, \ldots, x_n)$ is identically zero.

Lemma 4.6 ([Ore22; DL78; Zip79; Sch80]): Suppose that $f(x_1, \ldots, x_n)$ is an n-variate, degree d polynomial and $S \subseteq \mathbb{F}$ a finite subset. Choose $z_1, \ldots, z_n \xleftarrow{\$} S$ uniformly at random from S. Then $\mathbb{P}[\, f(z_1, \ldots, z_n) = 0\,] \leq \frac{d}{|S|}$.

Remark 4.7: Note that the bound in Lemma 4.6 is tight. Indeed choose any d distinct elements $s_1, \ldots, s_d$ from S. Let $f(x_1, \ldots, x_n) = \prod_{i=1}^{d}(x_1 - s_i)$. Then, uniformly randomly sample $z_1, \ldots, z_n$ from S. Then

$$\mathbb{P}[\, f(z_1, \ldots, z_n) = 0\,] = \mathbb{P}[\, z_1 \in \{s_1, \ldots, s_d\}\,] = \frac{d}{|S|}.$$

Lemma 4.6, known in the literature as the *Schwartz-Zippel Lemma*, gives an efficient randomized algorithm for polynomial identity testing by simply choosing $|S| > 2d$. If $|\mathbb{F}| \leq 2d$, it is common to allow the construction of a field extension for $\mathbb{F}$ and to evaluate $f(\overline{x})$ at this field extension.

One major question is to derandomize the Schwartz-Zippel lemma. Specifically, for some class of polynomials $\mathcal{C}$, such as polynomials whose minimal circuit size is $\leq s$, we would like an algorithm to determine if some $f \in \mathcal{C}$ is identically zero or not. It turns out that designing such an algorithm is equivalent to constructing *hitting sets*:

Definition 4.8 (Hitting Set Generator (HSG)): Let $\mathcal{C}$ be a class of n-variate polynomials over $\mathbb{F}$ We say that $\mathcal{G} \subseteq \mathbb{F}^n$ is a *hitting set generator*for $\mathcal{C}$ if for all $f \in \mathcal{C}$, f is identically zero if and only if $f(\overline{z}) = 0$ for some $\overline{z} \in \mathcal{H}$.

Closure results for ideals are useful for constructing hitting set generators, which are the algebraic-complexity analogs of pseudorandom generators and can be used to derandomize polynomial identity testing (PIT). The idea is as follows: for $s < n$, we seek a hitting set generator $G : \mathbb{F}^s \to \mathbb{F}^n$ against a family $\mathcal{C}$ of n-variate polynomials. This means that G has the property that for any nonzero polynomial $f \in \mathcal{C}$, we have $f \circ G \neq 0$. Here, one should think of $\mathcal{C}$ as a family of "low-complexity" polynomials. Let I be the ideal of polynomials f satisfying $f \circ G = 0$, also known as the *vanishing ideal* of G [HMM24].

To show that G has the desired hitting set property, it suffices to show that every nonzero $f \in I$ has sufficiently high complexity so that no such f can belong to $\mathcal{C}$. With a closure result for I, this reduces to proving lower bounds for a distinguished set of polynomials $\{g_1, \ldots, g_k\}$ for I. This can be viewed as a particular way of implementing the "hardness versus randomness" paradigm of [NW94; KI04] in the setting of algebraic complexity.

It turns out there is a deep connection between polynomial identity testing and factorization. It is shown in [KSS15] that if one can solve polynomial identity testing for the class $\mathcal{C}$ of size s, n-variate, degree d circuits over $\mathbb{Q}$ or $\mathbb{F}_{p^\ell}$ in time $\mathrm{poly}(s, n, d)$, then one can deterministically fully factorize circuits in $\mathcal{C}$ in polynomial time. Note that the converse direction is immediate to see. If one can factorize a polynomial in $\mathcal{C}$ in polynomial time, then one can determine if it is identically zero in polynomial time.

4.1.3. *More Than One Generator.* Recall the general question, Question 4.1, we want to study for ideals $I = \langle g_1, \ldots, g_k \rangle$. The principal ideal case discussed above corresponds to the complexity of $f(\overline{x}) \in I$ versus $g_1, \ldots, g_k$ for $k = 1$. What can we say about $k > 1$? As stated, this question is not necessarily well formed. Then a core issue is that for an expression $\sum_{i=1}^k f_i \cdot g_i \in I$, we have to somehow account for the possibility of cancellations which can take several hard-to-compute polynomials and combine them into an easy-to-compute polynomials.

Example 4.9: Suppose that $I = \left\langle \mathrm{perm}_n, \det_n - \mathrm{perm}_n \right\rangle$. Then it is conjectured that perm_n, and thus $\det_n - \mathrm{perm}_n$, requires a circuit of size exponential in n. However, $\det_n = \mathrm{perm}_n + (\det_n - \mathrm{perm}_n) \in I$ is known to have a circuit of size $\mathrm{poly}(n)$ computing it. This makes Question 4.1 have an answer which is not interesting. Adding $\det_n$ to the set of generators remedies this issue.

We see from Example 4.9 that to begin studying Question 4.1, one must first consider what the "right" set of generators is. For example, one could require that the generators we use satisfy some suitably nice algebraic properties. We can ask that they form a Gröbner basis or a regular sequence. In the various forms of determinantal ideals that we study, the generators we discuss will indeed form a Gröbner basis, and moreover they will be a linear-algebraic basis and enjoy nice combinatorial descriptions.

Conjecture 4.10 ([Gro20, Conjecture 6.3]): Let X be an $n \times n$ matrix of indeterminates and let I_n be the ideal generated by the $\frac{n}{2} \times \frac{n}{2}$ minors of X. Then for every nonzero polynomial $f \in I_n$, there is a constant-depth[‡] algebraic circuit of size $\mathrm{poly}(n)$ with f-oracle gates that computes the $m \times m$ determinant for some $m = n^{\Theta(1)}$.

[‡]The original statement of the conjecture does not specify constant-depth algebraic circuits. However, it is already known that the $m \times m$ determinant can be computed by algebraic circuits of size $\mathrm{poly}(m)$.

5. Current Work on Ideals with Straightening Laws

5.1. **Approximative Results.** Analogous to Bürgisser's theorem [Bür04] that the Factor Conjecture holds with respect to border complexity, Andrews and Forbes [AF22] proved that Grochow's conjecture holds with respect to border complexity. Formally, they proved the following theorem:

Theorem 5.1 ([AF22, Theorem 1.1]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{\substack{1\le i\le n\\ 1\le j\le m}}$ be an $n \times m$ matrix of variables over $\mathbb{F}$ and let $I^{\det}_{n,m,r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j}]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^2m^2\right)$ that approximately computes the $t \times t$ determinant for $t = \Theta\left(r^{1/3}\right)$.

Theorem 5.2 ([AF22, Theorem 1.1]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{1\le i,j\le 2n}$ be a $2n \times 2n$ matrix of variables over $\mathbb{F}$ and let $I^{\mathrm{pfaff}}_{2n,2r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j}]$ be the ideal generated by the Pfaffians of the $2r \times 2r$ principal submatrices of X. Let $f \in I^{\mathrm{pfaff}}_{2n,2r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^4\right)$ that approximately computes the $t \times t$ Pfaffian for $t = \Theta\left(r^{1/3}\right)$.

5.2. **Debordering.** In [DG25], we deborder the results of [AF22]. This is done via a novel application of the Isolation Lemma: Lemma 2.29 and Corollary 2.30.

Theorem 5.3 (Informal version of [DG25, Theorem 3.18, Corollary 3.19]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{\substack{1\le i\le n\\ 1\le j\le m}}$ be an $n \times m$ matrix of variables over $\mathbb{F}$. Let $I^{\det}_{n,m,r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j} \mid 1 \le i \le n,\ 1 \le j \le m]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then, there exists a depth-three f-oracle circuit of size $\mathrm{poly}(n, m, \deg(f))$ that computes the $t \times t$ determinant for $t = \Theta\left(r^{1/3}\right)$.

Theorem 5.4 (Informal version of [DG25, Theorem 4.10, Corollary 4.11]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{1\le i<j\le 2n}$ be a $2n \times 2n$ skew-symmetric matrix of variables over $\mathbb{F}$ so that $x_{j,i} = -x_{i,j}$ for $1 \le j < i \le n$ and $x_{i,i} = 0$ for $1 \le i \le 2n$. Let $I^{\mathrm{pfaff}}_{2n,2r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j} \mid 1 \le i < j \le 2n]$ be the ideal generated by the $2r \times 2r$ principal Pfaffians of X. Let $f \in I^{\mathrm{pfaff}}_{2n,2r}$ be a nonzero polynomial. Then, there exists a depth-three f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ Pfaffian of a skew-symmetric matrix for $t = \Theta\left(r^{1/3}\right)$.

5.3. Proof Overview for Determinantal Case.

5.3.1. *Linear Operators.* Our application of the determinantal straightening law (Corollary 3.6), together with the isolation lemma (Corollary 2.30) and extraction of coefficients (Lemma 2.27), allows us to isolate a canonical bitableau from the expansion of nonzero $f \in I^{\det}_{n,m,r}$ in the basis of standard bideterminants. In particular, we will show that this can be done via a polynomial-sized depth-three f-oracle circuit using the isolation lemma (Corollary 2.30) as well as coefficient extraction (Lemma 2.27). This is in contrast to the proof of [AF22], where they use a Kronecker-type substitution in order to isolate terms. To do this, we will give a more careful analysis of the terms that arise when applying the substitution operators to the expression of a polynomial $f(X) \in I^{\det}_{n,m,r}$.

Throughout, fix a field $\mathbb{F}$. Let n, m be natural numbers, and for $1 \le i \le n$ and $1 \le j \le m$, let $x_{i,j}$ be indeterminates. Throughout this section, let $X = (x_{i,j})$ be an $n \times m$ matrix of variables. Let $1 \le r \le \min(n, m)$ and define $I^{\det}_{n,m,r}$ to be the ideal in $\mathbb{F}[X]$ generated by the $r \times r$ minors of X.

Our first step will be to establish a series of linear transformations that send a conjugate semistandard bitableau to the anti-canonical conjugate semistandard bitableau of the same shape.

Definition 5.5 (Substitution operator): Let $i < j$. Define the operator $\mathrm{Sub}_{i\to j}$ which takes as input a tableau S and returns the tableau S' which is formed by taking each row of S which has an i but not a j, replacing that i with j, and then sorting the row in increasing order. We also define $h_i^j(S)$ to be the number of times i is replaced by j after applying $\mathrm{Sub}_{i\to j}$ to S.

Definition 5.6 (Canonical/anti-canonical tableau): Fix a natural number n. Let $\sigma = (\sigma_1 \ge \cdots \ge \sigma_k > 0)$ be a partition such that the length σ_i of each row is at most n. Define $K_{\sigma,n}$ to be the tableau of shape σ where row i has entries $1, 2, \ldots, \sigma_i$. Similarly, define $\overline{K}_{\sigma,n}$ to be the tableau of shape σ where row i has entries $(n - \sigma_i + 1, n - \sigma_i + 2, \ldots, n)$. We call $K_{\sigma,n}$ and $\overline{K}_{\sigma,n}$ the *canonical* and *anti-canonical* tableau respectively.[§] Note that $K_{\sigma,n}$ and $\overline{K}_{\sigma,n}$ are both conjugate semistandard. When n is clear from context, we will just write K_σ and $\overline{K}_\sigma$.

Example 5.7: If $\sigma = (5, 4, 2, 1)$ and $n = 7$, then K_σ and $\overline{K}_\sigma$ are given in Figure 1.

$$
\begin{array}{|c|c|c|c|c|}
\hline
1 & 2 & 3 & 4 & 5 \\
\hline
1 & 2 & 3 & 4 \\
\cline{1-4}
1 & 2 \\
\cline{1-2}
1 \\
\cline{1-1}
\end{array}
\qquad
\begin{array}{|c|c|c|c|c|}
\hline
3 & 4 & 5 & 6 & 7 \\
\hline
4 & 5 & 6 & 7 \\
\cline{1-4}
6 & 7 \\
\cline{1-2}
7 \\
\cline{1-1}
\end{array}
$$

FIGURE 1. The canonical and anti-canonical tableau for $\sigma = (5, 4, 2, 1)$ and $n = 7$.

Applying the substitution operators along the full lexicographic ordering transforms a conjugate semistandard tableau into the anti-canonical tableau of the same shape:

Lemma 5.8 ([dCEP80, stated before Corollary 1.7]): Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n. Then

$$(\mathrm{Sub}_{n-1\to n} \circ \mathrm{Sub}_{n-2\to n} \circ \mathrm{Sub}_{n-2\to n-1} \circ \cdots \circ \mathrm{Sub}_{2\to n} \circ \cdots \circ \mathrm{Sub}_{2\to 3} \circ \mathrm{Sub}_{1\to n} \circ \cdots \circ \mathrm{Sub}_{1\to 2})(S) = \overline{K}_\sigma.$$

[§] In some sources, such as [BV88], these are referred to as the *initial* and *final* tableau.

We now study how these substitution operators act on bideterminants $(S \mid T)(X)$. From properties of the determinant, we see that multiplying the matrix of variables X by a elementary matrix $E_{i,j}$ corresponds to applying substitution operators on S or T:

Lemma 5.9: Let λ be a new indeterminate. Let $(S \mid T)$ be a bitableau, not necessarily standard, of shape σ. For $0 \le h \le h_i^j(S)-1$, let $\mathcal{C}^h_{i\to j}(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing. Then we have that

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\mathrm{Sub}_{i\to j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S'\in\mathcal{C}^h_{i\to j}(S)} \pm(S' \mid T)(X). \tag{3}$$

Similarly, we also have that

$$(S \mid T)(XE_{i,j}(\lambda)^\top) = \pm\lambda^{h_i^j(T)}(S \mid \mathrm{Sub}_{i\to j}(T))(X) + \sum_{h=0}^{h_i^j(T)-1} \lambda^h \sum_{T'\in\mathcal{C}^h_{i\to j}(T)} \pm(S \mid T')(X). \tag{4}$$

Definition 5.10: In addition to the variables $x_{i,j}$, where $i \in [n]$ and $j \in [m]$, we introduce new sets of indeterminates $\Lambda = \{\lambda_{i,j} \mid 1 \le i < j \le n\}$ and $\Xi = \{\xi_{i,j} \mid 1 \le i < j \le m\}$. We have that $|\Lambda| = \binom{n}{2} = O\left(n^2\right)$ and $|\Xi| = \binom{m}{2} = O\left(m^2\right)$. The purpose of these new indeterminates is that they will allow us to keep track of the effects of substitution operators as we isolate a canonical bitableau.

Define $M \in \mathbb{F}[\Lambda]^{n\times n}$ and $N \in \mathbb{F}[\Xi]^{m\times m}$ by

$$\begin{aligned} M = {} & E_{1,2}(\lambda_{1,2})\cdots E_{1,n}(\lambda_{1,n})E_{2,3}(\lambda_{2,3})\cdots E_{2,n}(\lambda_{2,n}) \\ & \cdots E_{n-2,n-1}(\lambda_{n-2,n-1})E_{n-2,n}(\lambda_{n-2,n})E_{n-1,n}(\lambda_{n-1,n}) \end{aligned}$$

and

$$\begin{aligned} N = {} & E_{m-1,m}(\xi_{m-1,m})^\top E_{m-2,m}(\xi_{m-2,m})^\top E_{m-2,m-1}(\xi_{m-2,m-1})^\top \\ & \cdots E_{2,m}(\xi_{2,m})^\top \cdots E_{2,3}(\xi_{2,3})^\top E_{1,m}(\xi_{1,m})^\top \cdots E_{1,2}(\xi_{1,2})^\top. \end{aligned}$$

For an integer $k > 0$, let $J_k \in \mathbb{F}^{k\times k}$ be the $k \times k$ matrix with 1's along the anti-diagonal from the bottom left to the top right and 0's elsewhere. Note that $\det_k(J_k) = (-1)^{\binom{k}{2}} = \pm 1$. The substitution $X \mapsto J_n X J_m$ has the effect of replacing row i with row $n-i+1$ and column j with column $m-j+1$. This turns an anti-canonical bitableau $(\overline{K}_\sigma \mid \overline{K}_\sigma)$ into a canonical one $(K_\sigma \mid K_\sigma)$.

5.3.2. *Isolation of a Single Term.* We can repeatedly apply Lemma 5.9 and use Lemma 5.8 to send a bidterminant to an anticanonical one, plus some error terms. We describe these error terms so that we may later isolate them and get a circuit just for a summation of anticanonical bideterminants.

Corollary 5.11: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Then $f(MJ_nXJ_mN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k\in A} \widehat{c}_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell\in B} \widehat{c}_\ell \Lambda^{\overline{e}_\ell} \Xi^{\overline{f}_\ell} \cdot (S_\ell \mid T_\ell)(X), \tag{5}$$

such that the following hold:

(1) A is a nonempty finite set and B is finite set disjoint from A.

(2) All terms in Equation (5) have the form $c\Lambda^{\overline{e}}\Xi^{\overline{f}} \cdot (S \mid T)(X)$, where $c \in \mathbb{F}^\times$, $\overline{e} \in \{0,1,\dots,d\}^{\binom{[n]}{2}}$, $\overline{f} \in \{0,1,\dots,d\}^{\binom{[m]}{2}}$, and $(S \mid T)(X)$ is a bideterminant in $I^{\det}_{n,m,r}$ of degree d. In particular, if S and T have shape σ, then $|\sigma| \le d$.

(3) The triples $(\overline{e}_k, \overline{f}_k, \sigma_k)$ are distinct, where k ranges over A.

Next, we introduce variables $Y = \{y_1,\dots,y_n\}$ and $Z = \{z_1,\dots,z_m\}$. Define the diagonal matrices

$$D = \operatorname{diag}(y_1,\dots,y_n) \in \mathbb{F}[Y]^{n\times n} \quad \text{and} \quad D' = \operatorname{diag}(z_1,\dots,z_m) \in \mathbb{F}[Z]^{m\times m}.$$

Lemma 5.12: Let $(S \mid T)(X)$ be a bideterminant of degree at most d. Then

$$(S \mid T)(DXD') = Y^{\overline{s}} Z^{\overline{t}} \cdot (S \mid T)(X),$$

where $\overline{s} = (s_1,\dots,s_n) \in \{0,1,\dots,d\}^n$, $\overline{t} = (t_1,\dots,t_m) \in \{0,1,\dots,d\}^m$, s_i is the number of times i appears in S, t_j is the number of times j appears in T, i.e., $(s_1e_1 + \cdots + s_ne_n) \oplus (t_1e_1 + \cdots + t_me_m)$ is the multidegree of $(S \mid T)(X)$ with respect to the grading defined in Definition 3.8.

We will often simply say that $(S \mid T)(X)$ in Lemma 5.12 has multidegree $(\overline{s}, \overline{t})$ rather than writing it as $(s_1e_1 + \cdots + s_ne_n) \oplus (t_1e_1 + \cdots + t_me_m)$.

Next, we introduce another variable v, and modify D and D' as follows:

$$\widetilde{D} = \operatorname{diag}(y_1v, y_2v^2 \dots, y_nv^n) \in \mathbb{F}[Y,v]^{n\times n} \quad \text{and} \quad \widetilde{D}' = \operatorname{diag}(z_1v, z_2v^2, \dots, z_mv^m) \in \mathbb{F}[Z,v]^{m\times m}.$$

Lemma 5.13: Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(\widetilde{D}X\widetilde{D}') = Y^{\overline{s}} Z^{\overline{t}} v^{|S|+|T|} \cdot (S \mid T)(X),$$

where $(\overline{s}, \overline{t})$ is the multidegree of $(S \mid T)(X)$.

Lemma 5.14: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_n\widetilde{D}X\widetilde{D}'J_mN) \in \mathbb{F}[X,\Lambda,\Xi,Y,Z,v]$. View g as a univariate polynomial in v with coefficients in $\mathbb{F}[X,\Lambda,\Xi,Y,Z]$, and write $g = \sum_i \operatorname{coeff}_{v^i}(g)v^i$, where $\operatorname{coeff}_{v^i}(g) \in \mathbb{F}[X,\Lambda,\Xi,Y,Z]$ denotes the coefficient of v^i in g. Choose the smallest integer $d_{\min}$ such that $\operatorname{coeff}_{v^{d_{\min}}}(g) \ne 0$. Then $d_{\min} = 2|K_\sigma|$ for some shape σ with $|\sigma| \le d$

and we may write $\mathrm{coeff}_{\nu^{d_{\min}}}(g)$ as a finite sum

$$\mathrm{coeff}_{\nu^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} Y^{\overline{s}_k} Z^{\overline{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X), \tag{6}$$

such that the following hold:

(1) For each $k \in I$, $c_k \in \mathbb{F}^\times$, $(\sigma_k)_1 \geq r$, and $|\sigma_k| \leq d$.

(2) The tuples $(\overline{e}_k, \overline{f}_k, \overline{s}_k, \overline{t}_k)$ are distinct, where k ranges over I.

Lemma 5.14 helps us separate a collection of terms solely containing canonical bideterminants. The next lemma further applies the isolation lemma to single out one term from this collection.

Lemma 5.15: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_n\widetilde{D}X\widetilde{D}'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, \nu]$. Then there exist integers $z_t = O\big(d(n^2+m^2)\big)$ for each variable $t \in \Lambda \sqcup \Xi \sqcup Y \sqcup Z$, and an integer $z_\nu = O\big(d^2(n^2+m^2)\big)$ such that for the two variable substitution maps

$$\phi : t \mapsto w^{z_t} \qquad \text{for } t \in \Lambda \sqcup \Xi \sqcup Y \sqcup Z$$

and

$$\psi : \nu \mapsto w^{z_\nu},$$

we have that $h := (\psi \circ \phi)(g) \in \mathbb{F}[X, w]$ has $\deg_w(h) = O\big(d^3(n^3+m^3)\big)$, and that $\mathrm{coeff}_{w^i}(h) = c \cdot (K_\sigma \mid K_\sigma)(X)$ for some integer $i \leq \deg_w(h)$, where $c \in \mathbb{F}^\times$, $\sigma_1 \geq r$ and $|\sigma| \leq d$.

The next lemma further applies the isolation lemma to single out one term from this collection.

Theorem 5.16: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Assume $|\mathbb{F}| \geq c_0 d^3(n^3 + m^3)$, where $c_0 > 0$ is a large enough constant. Then, there exists a depth-three f-oracle circuit of size $O\big(n^2m^2d^3(n^3+m^3)\big) = \mathrm{poly}(n,m,d)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$. Furthermore, the top gate of this circuit is an addition gate, and the bottom layer consists of $O\big(nmd^3(n^3+m^3)\big)$ addition gates. The total number of gates and wires, excluding those between the bottom addition gates and the input gates, is $O\big(nmd^3(n^3+m^3)\big)$.

5.3.3. *Projection to the Determinant.* We have now seen that given an oracle to some nonzero $f(X) \in I^{\det}_{n,m,r}$, one can efficiently compute a canonical bideterminant efficiently with an f-oracle circuit. We now leverage the connection between algebraic branching programs and determinants. The remainder of the proof follows [AF22], which we include here for completeness. One key idea is exploiting the close connection between algebraic branching programs and determinants (Lemma 2.11 and Lemma 2.12).

We will use Theorems 5.16 to construct a constant-depth algebraic circuit of size $\mathrm{poly}(n, \deg(f))$ with f-oracle gates that computes the determinant. We start by proving the following more general theorem.

Theorem 5.17: Let $\mathbb{F}$ be a field of characteristic zero. Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g(y_1, \ldots, y_\ell) \in \mathbb{F}[\overline{y}]$ be a polynomial computable by an algebraic branching program with at most r vertices. Assume $|\mathbb{F}| \geq c_0 d^3(n^3 + m^3)$, where $c_0 > 0$ is a large enough constant. Then there is a depth-three f-oracle circuit Φ of size $O\left(nmd^4 \ell r(n^3 + m^3)\right)$ defined over $\mathbb{F}$ computing $g(\overline{y})$. Furthermore, the top layer of this circuit consists of an addition gate.

Proof: By Lemma 2.12, there is a homogeneous polynomial $\widehat{g}(\overline{y}, z) \in \mathbb{F}[\overline{y}, z]$ such that $\widehat{g}(\overline{y}, 1) = g(\overline{y})$ and $\widehat{g}(\overline{y}, z)$ can be computed by an algebraic branching program over $\mathbb{F}$ with at most r vertices. We may add isolated vertices such that the algebraic branching program has exactly r vertices. Then by Lemma 2.11, there is a matrix $A(\overline{y}, z) \in \mathbb{F}[\overline{y}, z]^{r \times r}$ such that

- $\det_r(A(\overline{y}, z)) = 1 + \widehat{g}(\overline{y}, z)$,
- $\det_i(A(\overline{y}, z)_{[i],[i]}) = 1$ for all $1 \leq i \leq r - 1$,

and all entries of $A(\overline{y}, z)$ have degree at most 1 in $\overline{y}$ and z. Extend $A(\overline{y}, z)$ to an $n \times m$ matrix by adding 1's to the main diagonal and 0's elsewhere.

By Theorems 5.16, there exists a depth-three f-oracle circuit C over $\mathbb{F}$ of size $O\left(n^2 m^2 d^3(n^3 + m^3)\right)$ computing $(K_\sigma \mid K_\sigma)(X)$ such that $\sigma_1 \geq r$ and $|\sigma| \leq d$. Furthermore, the top gate of C is an addition gate, the bottom layer of C consists of $O\left(nmd^3(n^3 + m^3)\right)$ addition gates, and the total number of gates and wires excluding the wires between the bottom addition gates and the input gates is $O\left(nmd^3(n^3 + m^3)\right)$. Replacing the nm input gates of C by the $O(\ell)$ new input gates $\overline{y}$, z, and 1, and further connecting these gates to the $O\left(nmd^3(n^3 + m^3)\right)$ bottom addition gates, we obtain a depth-three f-oracle circuit over $\mathbb{F}$ of size $O\left(nmd^3 \ell(n^3 + m^3)\right)$ computing

$$h(\overline{y}, z) := (K_\sigma \mid K_\sigma)(A(\overline{y}, z)) = (1 + \widehat{g}(\overline{y}, z))^t$$

where $t := |\{\, i \mid \sigma_i \geq r \,\}| \geq 1$. Note that $t \leq \widehat{\sigma}_1 \leq d$.

Let δ be a new indeterminate. Then under the map $y_i \mapsto \delta \cdot y_i$ and $z \mapsto \delta$, we have that

$$\begin{aligned} h(\delta \cdot \overline{y}, \delta) &= (1 + \widehat{g}(\delta \cdot \overline{y}, \delta))^t \\ &= (1 + \delta^{\deg(\widehat{g})} \widehat{g}(\overline{y}, 1))^t \\ &= (1 + \delta^{\deg(\widehat{g})} g(\overline{y}))^t \qquad (7) \\ &= \sum_{i=0}^{t} \binom{t}{i} \delta^{i \cdot \deg(\widehat{g})} g(\overline{y})^i = 1 + \delta^{\deg(\widehat{g})} g(\overline{y}) + \cdots . \end{aligned}$$

Then $\deg_\delta(h(\delta \cdot \overline{y}, \delta)) \leq t \cdot \deg(\widehat{g}) \leq dr$. For each $\alpha \in \mathbb{F}$, we can construct from the depth-three f-oracle circuit computing $h(\overline{y}, z)$ another f-oracle circuit C_α computing $h(\alpha\overline{y}, \alpha)$, whose size is $O\left(nmd^3\ell(n^3+m^3)\right)$ and top gate is an addition gate. Also, as $|\mathbb{F}| \geq c_0 d^3(n^3+m^3)$, where $c_0 > 0$ is a large enough constant, we may assume $|\mathbb{F}| \geq dr+1$. Therefore, by Lemma 2.27, we can compute $\mathrm{coeff}_{\delta^{\deg(\widehat{g})}}(h(\delta \cdot \overline{y}, \delta)) = g(\overline{y})$ using a depth-three f-oracle circuit of size

$$O\left(\deg_\delta(h(\delta \cdot \overline{y}, \delta)) \cdot nmd^3\ell(n^3+m^3)\right) = O\left(nmd^4\ell r(n^3+m^3)\right).$$

□

The following corollaries are a debordering of the results [AF22, Corollary 3.9, Corollary 3.10] of Andrews and Forbes. In particular, we can use Theorems 5.17 to construct a small constant-depth f-oracle circuit for the determinant, which partially resolves Conjecture 4.10 posed by Grochow as stated before.

Corollary 5.18: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $t = O\left(r^{1/3}\right)$ and let Y be a $t \times t$ matrix of indeterminates. Assume $|\mathbb{F}| \geq c_0 d^3(n^3+m^3)$, where $c_0 > 0$ is a large enough constant. Then there is a depth-three f-oracle circuit Φ of size $O\left(nmd^4rt^2(n^3+m^3)\right) \leq O\left(nmd^4r^{5/3}(n^3+m^3)\right)$ over $\mathbb{F}$ computing $\det_t(Y)$. Furthermore, the top gate of this circuit is an addition gate.

Proof: Mahajan and Vinay [MV97, Theorem 2] construct an algebraic branching program on $O\left(t^3\right) \leq r$ vertices which computes $\det_t(Y)$. Then, the total number of variables in Y is $t^2 = O\left(r^{2/3}\right)$. The result then follows by applying Theorems 5.17. □

6. Future Work and Open Problems

Andrews and Forbes [AF22] gave closure results for determinantal and Pfaffian ideals. Subsequent work [DG25] then debordered these ideal closure results. Both of these works relied heavily on the existence of a straightening law in order to control cancellations that may occur. A natural question is to ask if similar closure results can hold in other settings where straightening laws have been given.

6.1. **Symmetric Determinantal Ideals.**

Closure results have been given for the case of determinantal ideal as well as pfaffian ideals. A natural next step is to prove analogous results for the case of principal minors of a symmetric matrix.

Let $\mathbb{F}$ be a field. Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} := x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix of variables. For notational consistency with the prior subsections, we denote the polynomial ring $\mathbb{F}[x_{i,j} \mid 1 \leq i < j \leq 2n]$ by $\mathbb{F}[X]$.

Conjecture 6.1: Let $I_{n,r}^{\mathrm{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ principal minors of X. Let $f \in I_{n,r}^{\mathrm{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ Pfaffian of a skew-symmetric matrix for $t = \mathrm{poly}(n)$.

We give details of the straightening laws for the symmetric case. The notions of bitableau and bideterminants in this case are the same as in the determinantal case. We again have that the bideterminants span $\mathbb{F}[X]$ and a specific subset of standard ones will form an $\mathbb{F}$-basis. However, our notion of standard bideterminants will differ from the determinantal case.

Definition 6.2 (d-tableau, d-determinant): We say a bitableau $(S \mid T)$ of shape σ and its associated bideterminant $(S \mid T)(X)$ are *doubly-standard* if S and T are both conjugate semistandard Young tableau and if furthermore the single tableau formed by stacking alternating rows of S and T,

$$\begin{array}{|c|}\hline S_1 \\ \hline T_1 \\ \hline S_2 \\ \hline T_2 \\ \hline \vdots \\ \hline \vdots \\ \hline S_3 \\ \hline T_3 \\ \hline \end{array}$$

is conjugate semistandard. Following [Con94], we call such a doubly-standard bitableau a *d-tableau*. If $(S \mid T)$ is a d-tableau, we call $(S \mid T)(X)$ a *d-bideterminant*. We have that if $(S \mid T)(X)$ is a bideterminant of shape σ then $(S \mid T)(X)$ has total degree $|\sigma|$.

We denote the single tableau formed by stacking S and T by $S \oplus T$. The map $(S \mid T) \mapsto S \oplus T$ gives an injection from doubly-standard bitableau to conjugate semistandard tableau. Furthermore, the image of this map are conjugate semistandard bitableau whose shapes $\sigma = (\sigma_1 \geq \sigma_2 \geq \cdots \sigma_{2\ell})$ are such that for all $0 \leq i < \ell$ we have that $\sigma_{2i+1} = \sigma_{2(i+1)}$. Writing $\sigma = (\sigma_1 \geq \cdots \geq \sigma_\ell)$, define $2\sigma = (\sigma_1 = \sigma_1 \geq \sigma_2 = \sigma_2 \geq \cdots \geq \sigma_\ell = \sigma_\ell)$, i.e., 2σ is the shape of the partition formed by duplicating each row of the Young diagram for σ. Then this bijection is such that if $(S \mid T)$ is of shape σ then $S \oplus T$ is of shape 2σ. Finally, this bijection also preserves the lexicographic ordering in both sets of tableau. As such, we often identify $(S \mid T)$ with $S \oplus T$ as needed. An example of a doubly-standard bitableau is given in Figure 2. Note that $K_\sigma \oplus K_\sigma = K_{2\sigma}$ and $\overline{K}_\sigma \oplus \overline{K}_\sigma = \overline{K}_{2\sigma}$.

The following lemma shows that every monomial in $\mathbb{F}[X]$ can be expressed as a bideterminant, and hence the bideterminants span $\mathbb{F}[X]$. The proof is immediate from the definition.

$$(S \mid T) = \left(\begin{ytableau} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 7 \\ 5 & 6 \\ 7 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 \\ 6 & 7 \\ 7 \end{ytableau} \right) \longleftrightarrow \begin{ytableau} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 6 & 7 \\ 1 & 4 & 5 & 7 \\ 4 & 5 & 6 & 7 \\ 5 & 6 \\ 6 & 7 \\ 7 \\ 7 \end{ytableau} = S \oplus T.$$

FIGURE 2. A doubly-standard bitableau of shape $(5,4,2,1)$ shown as a bitableau as well as a single tableau of shape $(5,5,4,4,2,2,1,1)$.

Lemma 6.3: A degree d monomial $\prod_{i=1}^{d} x_{r_i,c_i}$ where for each $1 \leq i \leq d$, $r_i \leq c_i$ is given by a bideterminant:

$$\left(\begin{ytableau} r_1 \\ r_2 \\ \vdots \\ r_d \end{ytableau} \;\middle|\; \begin{ytableau} c_1 \\ c_2 \\ \vdots \\ c_d \end{ytableau} \right)(X) = \prod_{i=1}^{d} x_{r_i,c_i}.$$

As in the determinantal case, the symmetric determinantal case also admits a straightening law.

Theorem 6.4 ([dCP76, Theorem 5.1], [Con94, Theorem 2.2]): Let $(S \mid T)$ be a bitableau of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X),$$

where the $(A \mid B)(X)$ are d-bideterminants of shape $\tau \geq_{\text{lex}} \sigma$, and $c_{A,B} \in \mathbb{Z}$.

Lemma 6.3 and Theorems 6.4 together imply the following corollary.

Corollary 6.5: The d-bideterminants form an $\mathbb{F}$-basis of $\mathbb{F}[X]$. In particular, since bideterminants are homogeneous with respect to total degree, the degree-d component of $\mathbb{F}[X]$ has as basis the d-bideterminants of shape σ such that $|\sigma| = d$.

Let $I_{n,r}^{\text{sym}}$ be the ideal of $\mathbb{F}[X]$ generated by the $r \times r$ minors of X. We have the following crucial statement.

Proposition 6.6 ([Con94, Lemma 2.6]): Polynomials $f \in I_{n,r}^{\text{sym}}$ are supported by d-bideterminants of shape σ such that $\sigma_1 \geq r$.

Thus, all the tools needed for the symmetric case are given to prove the following in an analogous manner to the determinantal and Pfaffian cases:

Theorem 6.7: Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial, d the total degree of f, and $f(X) = \sum_{k\in[s]} \alpha_k(S_k \mid T_k)(X)$ the expansion of $f(X)$ in the d-bideterminant basis where $(S_k \mid T_k)$ is of shape

σ_k and each $\alpha_k \in \mathbb{F}$ is nonzero. Then there exists a partition σ and a depth-three f-oracle circuit of size $O\left(d^3 n^7\right)$ computing $\alpha[K_\sigma](X)$ for some nonzero $\alpha \in \mathbb{F}$.

The main obstacle in proving Conjecture 6.1 is thus giving an analogue of Lemma 2.11 for the symmetric setting. Valient's construction [Val79], connecting algebraic branching programs to determinantal complexity, gives a highly non-symmetric matrix. In particular, all upper-left corner proper minors are upper triangular with ones on the diagonal. This is not an issue in the skew-symmetric case.

Remark 6.8: Suppose that $g(\overline{y})$ can be computed by an r-vertex branching program. Let A be the $r \times r$ matrix such that $\det(A) = g(\overline{y})$ given by Lemma 2.11. Define B as the $2r \times 2r$ matrix $B := \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix}$. Then

$$\mathrm{pfaff}_{2r}(B) = \sqrt{\det(A)^2} = \sqrt{g(\overline{y})^2} = g(\overline{y}).$$

This is a key construction in the proofs of Theorems 5.2 and Theorems 5.4. However, we currently have no such analogous construction in the symmetric case.

6.2. **Algebras with Straightening Laws and Permanental Ideals.** The straightening laws for the determinantal, skew-symmetric, and even symmetric cases are examples of a more general construction defined by Eisenbud.

Definition 6.9 ([Eis80]): Suppose that A is a ring and $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that each $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R-algebra, and $H \subseteq A$ a finite poset which generates A as an R-algebra. Then A is an *Algebra with a Straightening Law* if

- A is a free R-module with a basis given by the standard monomials, and
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^{m} r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

  where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

For example, the ideals considered above $I^{\text{det}}_{n,m,r}$, $I^{\text{pfaff}}_{2n,2r}$, and $I^{\text{sym}}_{n,r}$ all form examples of algebras with a straightening law. The coordinate rings of the Grassmannian as well as flag varieties were also some of the first studied examples of algebras with straightening laws.

Studying the difference in computational power between the determinant and permanent polynomial is at the core of many questions in algebraic complexity. Thus, it is natural to consider the ideal $I^{\text{perm}}_{n,m,r}$ generated by $r \times r$ sub-permanents of an $n \times m$ matrix of variables. It is even known that $I^{\text{perm}}_{n,m,r}$ has a basis of standard bipermanents analogous to the determinantal case [Cla82]. However, it is not known if the coordinate ring has the structure of an algebra with a straightening law [GN25]. Furthermore, the proofs of Theorems 5.3, Theorems 5.4, and even in the symmetric case Theorems 6.7, all crucially rely on the fact that the determinant is alternating in the rows and columns. This property gives the crucial identity Lemma 5.9. However, the permanent is not alternating in the rows or columns of the matrix. Thus, we cannot expect that the same proof techniques as in [DG25] will give closure results for permanental ideals.

6.3. **Hankel Ideals.** Another case of ideals with a straightening law is the ideal generated by minors of a Hankel matrix. We follow [Con98] for the description of the standard monomials of the ideals of a Hankel matrix. We note that the description of standard monomials in the case of Hankel matrices differs to the case of standard and skew-symmetric matrices. Namely, we do not get the general structure of an algebra with a straightening law. Indeed, we will see that the product of two standard monomials need not be standard.

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \le i \le n$. Let X be the following matrix

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-1} & x_n \\ x_2 & x_3 & \cdots & \iddots & & \\ x_3 & \cdots & \iddots & & & \\ \vdots & \iddots & & & & \\ x_{n-1} & & & & & \\ x_n & & & & & \end{pmatrix}$$

For $1 \le j \le n$, let X_j be the following Hankel submatrix of X:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

The description of the standard monomials for the Hankel setting is more involved than the description for the standard determinantal or skew-symmetric pfaffian case. We begin with a description of how to index minors, as well as a useful term ordering.

Definition 6.10: For $a_1, \dots, a_s$ and $b_1, \dots, b_s$ such that $a_i + b_j - 1 \leq n$ for all $1 \leq i, j \leq s$, let $[a_1 \cdots a_s \mid b_1 \cdots b_s]$ denote the minor of X with rows indexed by the a_i's and columns indexed by the b_j's. A minor such that $a_i = i$ will be a *maximal minor*. We will denote maximal minors by only indexing their columns $[b_1 \cdots b_s]$

Let $>_L$ be the degree-lexicographic ordering on $\mathbb{F}[X]$ induced by $x_1 > \cdots > x_n$. For indices i, j, let $i <_1 j$ if and only if $i + 1 < j$. Then the $<_1$-chains $x_{a_1} \cdots x_{a_s}$, where $a_1 <_1 \cdots <_1 a_s$ are initial monomials of a minor of X under $>_L$.

We now describe a series of bijections which give a description of the standard monomials.

Definition 6.11: The $<_1$-chains are precisely the initial monomials of many minors of X, but they are the initial monomials of precisely one maximal minor. Thus, we have a bijection

$$\begin{aligned} \varphi \colon \{ <_1 -\text{chains of } \mathbb{F}[X] \} &\leftrightarrow \{ \text{maximal minors of } X \} \\ x_{a_1} \cdots x_{a_s} &\mapsto [a_1 \; a_2 - 1 \cdots a_s - s + 1]. \end{aligned}$$

Definition 6.12 (Canonical Decomposition): Let m be a monomial in the x_i's. Then we can decompose m into a product of $<_1$-chains in a "greedy" fashion by successively taking maximal $<_1$-chains and dividing them out from m until m is completely decomposed. We can represent this product of $<_1$-chains in a tableau. Call such a decomposition the *canonical decomposition* of m.

Definition 6.13: From the canonical decomposition, we get that φ induces an injective map

$$\begin{aligned} \Phi \colon \left\{ \text{monomials in the } x_i' \text{s} \right\} &\hookrightarrow \{ \text{products of maximal minors of } X \} \\ m &\mapsto \varphi(m_1) \cdots \varphi(m_k) \end{aligned}$$

where $m = m_1 \cdots m_k$ is the canonical decomposition of m.

Definition 6.14 (Standard Monomials): The *standard monomials* of $\mathbb{F}[\overline{x}]$ are the image of Φ. Note that we can describe the standard monomials via tableau, so that it makes sense to speak of the *shape* of a standard monomial.

For $1 \leq t \leq \min\{ j, n - j + 1 \}$ let $I_t(X_j)$ be the ideal generated by $t \times t$ minors of X_j.

Lemma 6.15 ([Con98, Corollary 2.2(b)]): For $t \leq j \leq n - t + 1$, we have that $I_t(X_j) = I_t(X_t)$. Thus, $I_t = I_t(X_j)$ only depends on t and n, not on j.

Note that products of standard monomials are not necessarily standard.

Example 6.16: Let $\mu_1 = [1,2]$, $\mu_2 = [2]$, and $\mu_3 = [2,4]$. Then $\mu_1 \cdot \mu_2$ has initial monomial $x_1x_2x_3$ which corresponds to the standard monomial $[1,2][2] = \mu_1 \cdot \mu_2$. Similarly, $\mu_2 \cdot \mu_3$ has initial monomial $x_2^2x_5$ which corresponds to the standard monomial $[2,4][2] = \mu_2 \cdot \mu_3$. However, $\mu_1 \cdot \mu_2 \cdot \mu_3$ has initial monomial $x_1x_2^2x_3x_5$ which corresponds to $[1,2,3][2]^2$ which differs from $[1,2][2][2,4]$.

However, we can still get a basis of I_t via maximal minors.

Theorem 6.17 ([Con98, Corollary 2.2(a, c)]): For $j > t$, every t-minor of X_j is a linear combination of t-minors of X_{j-1}. Furthermore, every t-minor is a linear combination of maximal t-minors.

Furthermore, since variables are shared between rows of a minor, it is hard to describe how linear transformations act on X. Specifically, we cannot describe $E_{i,j}X$ just by substituting variables. Thus, much of the proof strategy of [DG25] does not apply to the case of Hankel minor ideals. Specifically, we cannot simulate an increase or decrease in entries of a bitableau by applying a linear change of variables. However, we have the following identity:

Lemma 6.18: For $a_1,\dots,a_s$ and $b_1,\dots,b_s$ such that $a_i + b_j - 1 \le n$ for all $1 \le i, j \le s$, let $[a_1 \cdots a_s \mid b_1 \cdots b_s]$ denote the minor of X with rows indexed by the a_i's and columns indexed by the b_j's. Then

$$\sum_{i=2}^{n} x_{i-1}\frac{\partial}{\partial x_i}[a_1 \cdots a_s \mid b_1 \cdots b_s] = \sum_{i=1}^{n}[a_1 \cdots a_s \mid b_1 \cdots b_i - 1 \cdots b_s].$$

Thus, we can use a sequence of partial derivative operators to try to transform a standard monomial to a monomial describing a minor in the upper-left corner of X.

This approach has a couple of issues. Consider a 1 row bitableau in the determinantal case which indexes a minor in the upper left corner. Then the substitution operators $\mathrm{Sub}_{i\to j}$ of [dCEP80], used in the analysis of [AF22; DG25], applied to this bitableau will leave the bitableau unchanged. In other words, the substitution operators leave certain bitableau unchanged. However, the shift operators of Lemma 6.18 do not have this property. Namely, the indices of a given standard monomial are allowed to be nonpositive integers where $x_i = 0$ for $i \le 0$. Moreover, computing partial derivatives is hard. A theorem of Baur and Strassen [BS83] states that single partial derivatives are not too hard to compute

Proposition 6.19 ([BS83]): Let $f(x_1,\dots,x_n)$ be a polynomial that has a circuit C of size s, depth d, and fan-in k. Then, there exists a circuit D of size $O(sk)$, depth $O(d)$, and fan-in k simultaneously computing $\partial_1 f(\overline{x}),\dots,\partial_n f(\overline{x})$.

However, computing higher order partial derivatives will in general give an exponential blowup in circuit size. For example, computing second order partial derivatives efficiently will give faster matrix multiplication algorithms than are currently known. In particular, suppose that given a circuit of size s computing $f(x_1,\dots,x_n)$ one can show that a circuit computing $\frac{\partial^2 f}{\partial x_i \partial x_j}$ has size $O\left(s+n^2\right)$. Then, one can multiply matrices in $O\left(n^2\right)$ [CKW11].

References

[AF22] Robert Andrews and Michael A. Forbes. *Ideals, Determinants, and Straightening: Proving and Using Lower Bounds for Polynomial Ideals*. Nov. 27, 2022. DOI: 10.48550/arXiv.2112.00792. arXiv: 2112.00792 [cs.CC]. Pre-published.

[AV08] Manindra Agrawal and V. Vinay. "Arithmetic Circuits: A Chasm at Depth Four". In: *2008 49th Annual IEEE Symposium on Foundations of Computer Science*. 2008 49th Annual IEEE Symposium on Foundations of Computer Science. Oct. 2008, pp. 67–75. DOI: 10.1109/FOCS.2008.32.

[BC03] Winfried Bruns and Aldo Conca. "Gröbner Bases and Determinantal Ideals". In: *Commutative Algebra, Singularities and Computer Algebra*. Springer Netherlands, 2003, pp. 9–66. DOI: 10.1007/978-94-007-1092-4_2.

[BDS26] C. S. Bhargav, Prateek Dwivedi, and Nitin Saxena. *A Primer on the Closure of Algebraic Complexity Classes under Factoring*. Version 2. June 9, 2026. DOI: 10.48550/arXiv.2506.19604. arXiv: 2506.19604 [cs.CC]. Pre-published.

[BS83] Walter Baur and Volker Strassen. "The Complexity of Partial Derivatives". In: *Theoretical Computer Science* 22.3 (Feb. 1, 1983), pp. 317–330. ISSN: 0304-3975. DOI: 10.1016/0304-3975(83)90110-X.

[Bür00] Peter Bürgisser. *Completeness and Reduction in Algebraic Complexity Theory*. Springer Berlin Heidelberg, 2000. ISBN: 978-3-662-04179-6. DOI: 10.1007/978-3-662-04179-6.

[Bür04] Peter Bürgisser. "The Complexity of Factors of Multivariate Polynomials". In: *Foundations of Computational Mathematics* 4.4 (Nov. 2004), pp. 369–396. ISSN: 1615-3375, 1615-3383. DOI: 10.1007/s10208-002-0059-5.

[Bür24] Peter Bürgisser. *Completeness Classes in Algebraic Complexity Theory*. June 10, 2024. DOI: 10.48550/arXiv.2406.06217. arXiv: 2406.06217 [cs]. Pre-published.

[BV88] Winfried Bruns and Udo Vetter. *Determinantal Rings*. Springer, 1988. DOI: 10.1007/BFb0080378.

[CKW11] Xi Chen, Neeraj Kayal, and Avi Wigderson. "Partial Derivatives in Arithmetic Complexity and Beyond". In: *Found. Trends Theor. Comput. Sci.* 6.1–2 (2011), pp. 1–138. DOI: 10.1561/0400000043.

[Cla82] Michael Clausen. "A Straightening Formula for Bipermanents". In: *Linear and Multilinear Algebra* 11.1 (Jan. 1, 1982), pp. 33–38. ISSN: 0308-1087. DOI: 10.1080/03081088208817430.

[Con94] Aldo Conca. “Gröbner bases of ideals of minors of a symmetric matrix”. In: *Journal of Algebra* 166.2 (1994), pp. 406–421. DOI: 10.1006/jabr.1994.1160.

[Con98] Aldo Conca. “Straightening Law and Powers of Determinantal Ideals of Hankel Matrices”. In: *Advances in Mathematics* 138.2 (Sept. 25, 1998), pp. 263–292. ISSN: 0001-8708. DOI: 10.1006/aima.1998.1740.

[Csa76] L. Csanky. “Fast Parallel Matrix Inversion Algorithms”. In: *SIAM Journal on Computing* 5.4 (1976), pp. 618–623. DOI: 10.1137/0205040. eprint: https://doi.org/10.1137/0205040.

[dCEP80] Corrado de Concini, David Eisenbud, and Claudio Procesi. “Young Diagrams and Determinantal Varieties”. In: *Inventiones mathematicae* 56.2 (1980), pp. 129–165. DOI: 10.1007/BF01392548.

[dCP76] Corrado de Concini and Claudio Procesi. “A Characteristic Free Approach to Invariant Theory”. In: *Advances in Mathematics* 21.3 (1976), pp. 330–354. ISSN: 0001-8708. DOI: 10.1016/S0001-8708(76)80003-5.

[DG25] Anakin Dey and Zeyu Guo. *Debordering Closure Results in Determinantal and Pfaffian Ideals*. Nov. 21, 2025. Electronic Colloquium on Computational Complexity: TR25-189. URL: https://eccc.weizmann.ac.il/report/2025/189/ (visited on 03/24/2026). Pre-published.

[DL25] Pranjal Dutta and Vladimir Lysikov. *Recent Advances in Debordering Methods*. Oct. 15, 2025. DOI: 10.48550/arXiv.2510.13049. arXiv: 2510.13049 [cs]. Pre-published.

[DL78] Richard A. Demillo and Richard J. Lipton. “A Probabilistic Remark on Algebraic Program Testing”. In: *Information Processing Letters* 7.4 (June 1, 1978), pp. 193–195. ISSN: 0020-0190. DOI: 10.1016/0020-0190(78)90067-4.

[DRS74] Peter Doubilet, Gian-Carlo Rota, and Joel Stein. “On the Foundations of Combinatorial Theory: IX Combinatorial Methods in Invariant Theory”. In: *Studies in Applied Mathematics* 53.3 (1974), pp. 185–216. DOI: 10.1002/sapm1974533185.

[Eis80] David Eisenbud. “Introduction to Algebras with Straightening Laws”. In: *Ring Theory and Algebra III*. 3rd J. Clarence Karcher Algebra and Ring Theory Conference (Oklahoma). Vol. 55. New York and Basel: Marcel Dekker, Inc., 1980, pp. 243–268. URL: https://eisenbud.github.io/papers/pdfs/1980-001.pdf.

[For24] Michael A. Forbes. “Low-Depth Algebraic Circuit Lower Bounds over Any Field”. In: *39th Computational Complexity Conference (CCC 2024)*. Ed. by Rahul Santhanam. Vol. 300. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 31:1–31:16. ISBN: 978-3-95977-331-7. DOI: 10.4230/LIPIcs.CCC.2024.31.

[Ful96] William Fulton. *Young Tableaux: With Applications to Representation Theory and Geometry*. 2nd ed. London Mathematical Society Student Texts 35. Cambridge: Cambridge University Press, 1996.

[GN25] Joshua A. Grochow and Abhiram Natarajan. *Grobner Bases Native to Term-ordered Commutative Algebras, with Application to the Hodge Algebra of Minors*. Oct. 13, 2025. URL: https://arxiv.org/abs/2510.11212v1 (visited on 11/11/2025). Pre-published.

[Gro] Joshua Grochow. http://cstheory.stackexchange.com/questions/19261/degree-restriction-for-polynomials-in-mathsfvp/19268.

[Gro20] Joshua A. Grochow. "Complexity in Ideals of Polynomials: Questions on Algebraic Complexity of Circuits and Proofs". In: *Bull. EATCS* 131 (2020). URL: http://eatcs.org/beatcs/index.php/beatcs/article/view/620.

[Gup+13] Ankit Gupta, Pritish Kamath, Neeraj Kayal, and Ramprasad Saptharishi. *Arithmetic Circuits: A Chasm at Depth Three*. Feb. 11, 2013. Electronic Colloquium on Computational Complexity: TR13-026. URL: https://eccc.weizmann.ac.il/report/2013/026/. Pre-published.

[Har92] Joe Harris. *Algebraic Geometry*. Vol. 133. Graduate Texts in Mathematics. New York, NY: Springer, 1992. ISBN: 978-1-4419-3099-6 978-1-4757-2189-8. DOI: 10.1007/978-1-4757-2189-8.

[HMM24] Ivan Hu, Dieter van Melkebeek, and Andrew Morgan. "Polynomial Identity Testing via Evaluation of Rational Functions". In: *Theory of Computing* 20.1 (July 18, 2024), pp. 1–70. DOI: 10.4086/toc.2024.v020a001.

[Hod43] W. V. D. Hodge. "Some Enumerative Results in the Theory of Forms". In: *Mathematical Proceedings of the Cambridge Philosophical Society* 39.1 (Mar. 1943), pp. 22–30. DOI: 10.1017/S0305004100017631.

[HT92] Jürgen Herzog and Ngô Viêt Trung. "Gröbner Bases and Multiplicity of Determinantal and Pfaffian Ideals". In: *Advances in Mathematics* 96.1 (1992), pp. 1–37. DOI: 10.1016/0001-8708(92)90050-U.

[HY11] Pavel Hrubeš and Amir Yehudayoff. "Arithmetic Complexity in Ring Extensions". In: *Theory of Computing* 7.8 (May 9, 2011), pp. 119–129. DOI: 10.4086/toc.2011.v007a008.

[Kal87] E. Kaltofen. "Single-Factor Hensel Lifting and Its Application to the Straight-Line Complexity of Certain Polynomials". In: *Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing*. STOC '87. New York, NY, USA: Association for Computing Machinery, Jan. 1, 1987, pp. 443–452. ISBN: 978-0-89791-221-1. DOI: 10.1145/28395.28443.

[KI04] Valentine Kabanets and Russell Impagliazzo. "Derandomizing Polynomial Identity Tests Means Proving Circuit Lower Bounds". In: *computational complexity* 13.1–2 (Dec. 2004), pp. 1–46. ISSN: 1016-3328, 1420-8954. DOI: 10.1007/s00037-004-0182-6.

[Koi12] Pascal Koiran. *Arithmetic Circuits: The Chasm at Depth Four Gets Wider*. Mar. 23, 2012. DOI: 10.48550/arXiv.1006.4700. arXiv: 1006.4700 [cs.CC]. Pre-published.

[Kro82] L. Kronecker. “Grundzüge Einer Arithmetischen Theorie Der Algebraische Grössen.” ger. In: *Journal für die reine und angewandte Mathematik* 1882.92 (1882), pp. 1–122. DOI: 10.1515/crll.1882.92.1.

[KS01] Adam R. Klivans and Daniel Spielman. “Randomness Efficient Identity Testing of Multivariate Polynomials”. In: *Proceedings of the Thirty-Third Annual ACM Symposium on Theory of Computing*. STOC01: 33rd ACM Symposium on Theory of Computing. Hersonissos Greece: ACM, July 6, 2001, pp. 216–223. ISBN: 978-1-58113-349-3. DOI: 10.1145/380752.380801.

[KSS15] Swastik Kopparty, Shubhangi Saraf, and Amir Shpilka. “Equivalence of Polynomial Identity Testing and Polynomial Factorization”. In: *computational complexity* 24.2 (June 1, 2015), pp. 295–331. ISSN: 1420-8954. DOI: 10.1007/s00037-015-0102-y.

[LB18] V. Lakshmibai and Justin Brown. *Flag Varieties*. Vol. 53. Texts and Readings in Mathematics. Singapore: Springer, 2018. ISBN: 978-981-13-1393-6. DOI: 10.1007/978-981-13-1393-6.

[LST25] Nutan Limaye, Srikanth Srinivasan, and Sébastien Tavenas. “Superpolynomial Lower Bounds Against Low-Depth Algebraic Circuits”. In: *Journal of the ACM* 72.4 (July 24, 2025), 26:1–26:35. ISSN: 0004-5411. DOI: 10.1145/3734215.

[MV97] Meena Mahajan and V. Vinay. *Determinant: Combinatorics, Algorithms, and Complexity*. TR97-036. Electronic Colloquium on Computational Complexity (ECCC), Sept. 10, 1997. URL: https://eccc.weizmann.ac.il/eccc-reports/1997/TR97-036/index.html.

[MVV87] Ketan Mulmuley, Umesh V. Vazirani, and Vijay V. Vazirani. “Matching is as Easy as Matrix Inversion”. In: *Combinatorica* 7.1 (1987), pp. 105–113. DOI: 10.1007/BF02579206.

[NW94] Noam Nisan and Avi Wigderson. “Hardness vs Randomness”. In: *Journal of Computer and System Sciences* 49.2 (Oct. 1, 1994), pp. 149–167. ISSN: 0022-0000. DOI: 10.1016/S0022-0000(05)80043-1.

[Old40] Rufus Oldenburger. “Polynomials in Several Variables”. In: *Annals of Mathematics* 41.3 (1940), pp. 694–710. ISSN: 0003-486X. DOI: 10.2307/1968741. JSTOR: 1968741.

[Ore22] Øystein Ore. “Über Höhere Kongruenzen”. In: (1922), p. 15.

[Ost54] A. Ostrowski. “On Two Problems in Abstract Algebra Connected with Horner’s Rule”. In: *Studies in Mathematics and Mechanics*. Academic Press, 1954, pp. 40–48. ISBN: 978-1-4832-3272-0. URL: http://dx.doi.org/10.1016/B978-1-4832-3272-0.50010-7.

[Pal06] Francesco Palatini. “Sulle superficie algebriche i cui S_h $(h+1)$-seganti non riempiono lo spazio ambiente”. In: *Atti della R. Acc. delle Scienze di Torino* 41 (1906), pp. 634–640. URL: https://archive.org/details/attidellarealeac41real/page/634.

[Pan66] V Ya Pan. "Methods of Computing Values of Polynomials". In: *Russian Mathematical Surveys* 21.1 (Feb. 1966), pp. 105–136.

[Rai] Claudiu Raicu. *Algebras with Straightening Law*. URL: https://www.slmath.org/ckeditor_assets/attachments/783/raicu.pdf.

[Sap21] Ramprasad Saptharishi. *A Survey of Lower Bounds in Arithmetic Circuit Complexity*. July 2021. URL: https://github.com/dasarpmar/lowerbounds-survey/releases.

[Sch80] J. T. Schwartz. "Fast Probabilistic Algorithms for Verification of Polynomial Identities". In: *Journal of the ACM (JACM)* 27.4 (Oct. 1, 1980), pp. 701–717. ISSN: 0004-5411. DOI: 10.1145/322217.322225.

[Shp16] Amir Shpilka. *Some Accessible Open Problems*. 2016. URL: https://www.cs.tau.ac.il/~shpilka/wact2016/concreteOpenProblems/openprobs.pdf.

[Str69] Volker Strassen. "Gaussian Elimination Is Not Optimal". In: *Numerische Mathematik* 13.4 (Aug. 1, 1969), pp. 354–356. ISSN: 0945-3245. DOI: 10.1007/BF02165411.

[Str73] Volker Strassen. "Vermeidung von divisionen." In: *Journal für die reine und angewandte Mathematik* 264 (1973), pp. 184–202. URL: http://eudml.org/doc/151394.

[Stu08] Bernd Sturmfels. *Algorithms in Invariant Theory*. 2nd ed. Texts & Monographs in Symbolic Computation. Springer Vienna, 2008, p. vii 197. 197 pp. ISBN: 978-3-211-77417-5. DOI: 10.1007/978-3-211-77417-5.

[Ter11] Alessandro Terracini. "Sulle V_k per cui la varietà degli S_h $(h+1)$-seganti ha dimensione minore dell'ordinario". In: *Rend. Circ. Mat.* 31 (1911), pp. 392–396. DOI: 10.1007/BF03018812.

[Tut47] William T Tutte. "The factorization of linear graphs". In: *Journal of the London Mathematical Society* 1.2 (1947), pp. 107–111.

[Val79] L. G. Valiant. "Completeness Classes in Algebra". In: *Proceedings of the Eleventh Annual ACM Symposium on Theory of Computing*. STOC '79. New York, NY, USA: Association for Computing Machinery, Apr. 30, 1979, pp. 249–261. ISBN: 978-1-4503-7438-5. DOI: 10.1145/800135.804419.

[Val81] L. G. Valiant. "Reducibility by Algebraic Projections". In: 28 (Apr. 25, 1981). DOI: 10.5169/SEALS-52240.

[You28] Alfred Young. "On Quantitative Substitutional Analysis". In: *Proceedings of the London Mathematical Society* s2-28.1 (Jan. 1928), pp. 255–292. ISSN: 0024-6115. DOI: 10.1112/plms/s2-28.1.255.

[Zip79] Richard Zippel. "Probabilistic Algorithms for Sparse Polynomials". In: *Symbolic and Algebraic Computation*. Ed. by Edward W. Ng. Berlin, Heidelberg: Springer, 1979, pp. 216–226. ISBN: 978-3-540-35128-3. DOI: 10.1007/3-540-09519-5_73.